

# EDA For Datasets

Group 5

2026-09-08

## Table of contents

|                                                                                                       |           |
|-------------------------------------------------------------------------------------------------------|-----------|
| <b>Install packages if they are not installed:</b>                                                    | <b>3</b>  |
| Loading the packages we need: . . . . .                                                               | 3         |
| <b>This is the file path variable for the data set folder:</b>                                        | <b>3</b>  |
| <b>Creating the data frames for all the data sets:</b>                                                | <b>3</b>  |
| Finding the one or more parsing issues: . . . . .                                                     | 3         |
| We see that Data set 11 has a few problems in column 30, lets inspect what<br>column it is: . . . . . | 5         |
| <b>Quick summary about column and row counts for all datasets</b>                                     | <b>6</b>  |
| <b>Exploring the data:</b>                                                                            | <b>6</b>  |
| <b>Check For missing Data:</b>                                                                        | <b>29</b> |
| Total missing values for each dataset . . . . .                                                       | 29        |
| Diving deeper into which columns have the missing values in each dataset . . .                        | 30        |
| <b>Checking for duplicate Rows:</b>                                                                   | <b>38</b> |
| Total duplicated values for each dataset . . . . .                                                    | 38        |
| Diving deeper into the duplicated values in each dataset . . . . .                                    | 40        |
| <b>Which Datasets we choose:</b>                                                                      | <b>46</b> |
| <b>Which Datasets we dont choose:</b>                                                                 | <b>46</b> |

|                                                                                                                                                                                              |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Check intersect comparison to see if the datasets are compatible with each other:</b>                                                                                                     | <b>47</b> |
| Check the household ID matching % . . . . .                                                                                                                                                  | 48        |
| For person level checks we need to use composite keys so we first merge the<br>household_id and person_id into a new id variable and then compare<br>those values across datasets: . . . . . | 48        |
| <b>Analysis of our chosen datasets:</b>                                                                                                                                                      | <b>48</b> |
| Check for Empty string columns . . . . .                                                                                                                                                     | 48        |
| Check for Negative values in age . . . . .                                                                                                                                                   | 49        |
| Check for invalid values in the columns . . . . .                                                                                                                                            | 51        |
| Dataset 1: . . . . .                                                                                                                                                                         | 51        |
| Dataset 2: . . . . .                                                                                                                                                                         | 52        |
| Dataset 3: . . . . .                                                                                                                                                                         | 55        |
| Dataset 8: . . . . .                                                                                                                                                                         | 60        |
| <b>Graphs for our chosen datasets:</b>                                                                                                                                                       | <b>66</b> |
| Dataset 1 Graphs: . . . . .                                                                                                                                                                  | 66        |
| Summary of D01 Ages . . . . .                                                                                                                                                                | 67        |
| Province Distribution . . . . .                                                                                                                                                              | 67        |
| Finding Problem Province Naming . . . . .                                                                                                                                                    | 68        |
| Fixing Provinces . . . . .                                                                                                                                                                   | 69        |
| Fixed Provinces . . . . .                                                                                                                                                                    | 69        |
| Province Distribution . . . . .                                                                                                                                                              | 69        |
| Dataset 2 Graphs: . . . . .                                                                                                                                                                  | 70        |
| Showing Most Frequent Education Code . . . . .                                                                                                                                               | 71        |
| Showing All Education Code Counts . . . . .                                                                                                                                                  | 71        |
| Dataset 3 Graphs: . . . . .                                                                                                                                                                  | 73        |
| Finding Most Frequent Employment Code . . . . .                                                                                                                                              | 73        |
| Showing Employment Codes . . . . .                                                                                                                                                           | 74        |
| Plotting Salaries . . . . .                                                                                                                                                                  | 74        |
| Salary Summary . . . . .                                                                                                                                                                     | 75        |
| Finding Unusable Salary Rows . . . . .                                                                                                                                                       | 75        |
| Finding Unusable Salary Row % . . . . .                                                                                                                                                      | 76        |
| Fixed Salary Graph . . . . .                                                                                                                                                                 | 76        |
| Dataset 8 Graphs: . . . . .                                                                                                                                                                  | 77        |
| Mobile Internet Access . . . . .                                                                                                                                                             | 78        |
| Household Smartphones . . . . .                                                                                                                                                              | 79        |
| <b>What to handle in preprocessing:</b>                                                                                                                                                      | <b>80</b> |
| <b>Initial Hypotheses:</b>                                                                                                                                                                   | <b>81</b> |

## Install packages if they are not installed:

```
#install.packages(c("tidyverse", "skimr", "janitor", "visdat"))
```

## Loading the packages we need:

```
library(tidyverse)
library(skimr)
library(janitor)
library(visdat)
```

## This is the file path variable for the data set folder:

```
directory <- "../Datasets/"
```

## Creating the data frames for all the data sets:

```
D01 <- read_csv(file.path(directory, "D01_Population_Profile.csv"), show_col_types = FALSE, col_types = NULL)
D02 <- read_csv(file.path(directory, "D02_Learning_Profile.csv"), show_col_types = FALSE, col_types = NULL)
D03 <- read_csv(file.path(directory, "D03_Economic_Participation.csv"), show_col_types = FALSE, col_types = NULL)
D04 <- read_csv(file.path(directory, "D04_Dwelling_Conditions.csv"), show_col_types = FALSE, col_types = NULL)
D05 <- read_csv(file.path(directory, "D05_Basic_Services.csv"), show_col_types = FALSE, col_types = NULL)
D06 <- read_csv(file.path(directory, "D06_Household_Energy.csv"), show_col_types = FALSE, col_types = NULL)
D07 <- read_csv(file.path(directory, "D07_Household_Wellbeing.csv"), show_col_types = FALSE, col_types = NULL)
D08 <- read_csv(file.path(directory, "D08_Connectivity_Access.csv"), show_col_types = FALSE, col_types = NULL)
D09 <- read_csv(file.path(directory, "D09_Household_Resources.csv"), show_col_types = FALSE, col_types = NULL)
D10 <- read_csv(file.path(directory, "D10_Mobility_Access.csv"), show_col_types = FALSE, col_types = NULL)
D11 <- read_csv(file.path(directory, "D11_Labour_Market.csv"), show_col_types = FALSE, col_types = NULL)
```

## Finding the one or more parsing issues:

```
problems(D01)
```

```
# A tibble: 0 x 5  
# i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

```
problems(D02)
```

```
# A tibble: 0 x 5  
# i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

```
problems(D03)
```

```
# A tibble: 0 x 5  
# i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

```
problems(D04)
```

```
# A tibble: 0 x 5  
# i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

```
problems(D05)
```

```
# A tibble: 0 x 5  
# i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

```
problems(D06)
```

```
# A tibble: 0 x 5  
# i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

```
problems(D07)
```

```
# A tibble: 0 x 5  
# i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

```
problems(D08)
```

```
# A tibble: 0 x 5  
# i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

```
problems(D09)
```

```
# A tibble: 0 x 5  
# i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

```
problems(D10)
```

```
# A tibble: 0 x 5  
# i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

```
problems(D11)
```

```
# A tibble: 45 x 5  
   row   col expected actual      file  
   <int> <int> <chr>    <chr>    <chr>  
1  3044   30 a double not recorded D:/Github/BIN381/Datasets/D11_Labour_Marke~  
2  7201   30 a double Unknown    D:/Github/BIN381/Datasets/D11_Labour_Marke~  
3  7336   30 a double not recorded D:/Github/BIN381/Datasets/D11_Labour_Marke~  
4  7859   30 a double N/A        D:/Github/BIN381/Datasets/D11_Labour_Marke~  
5  9505   30 a double N/A        D:/Github/BIN381/Datasets/D11_Labour_Marke~  
6 12565   30 a double Unknown    D:/Github/BIN381/Datasets/D11_Labour_Marke~  
7 13217   30 a double N/A        D:/Github/BIN381/Datasets/D11_Labour_Marke~  
8 14545   30 a double N/A        D:/Github/BIN381/Datasets/D11_Labour_Marke~  
9 15029   30 a double not recorded D:/Github/BIN381/Datasets/D11_Labour_Marke~  
10 17084   30 a double Unknown    D:/Github/BIN381/Datasets/D11_Labour_Marke~  
# i 35 more rows
```

**We see that Data set 11 has a few problems in column 30, lets inspect what column it is:**

```
names(D11)[30] #Gives column name of col 30
```

```
[1] "hours_worked_raw"
```

```
nrow(problems(D11)) #Shows how many problems there are
```

```
[1] 45
```

This tells us that 45 rows in the hours\_worked\_raw column has issues in data set 11 that need to be resolved in the pre-processing

## Quick summary about column and row counts for all datasets

```
#Tibble creates a small dataframe from scratch modern way of doing data.frame
tibble(
  dataset = c("D01","D02","D03","D04","D05","D06","D07","D08","D09","D10","D11"),
  rows = c(nrow(D01),nrow(D02),nrow(D03),nrow(D04),nrow(D05),nrow(D06),nrow(D07),nrow(D08),nrow(D09),nrow(D10),nrow(D11)),
  cols = c(ncol(D01),ncol(D02),ncol(D03),ncol(D04),ncol(D05),ncol(D06),ncol(D07),ncol(D08),ncol(D09),ncol(D10),ncol(D11))
)
```

```
# A tibble: 11 x 3
  dataset  rows  cols
  <chr>    <int> <int>
1 D01      70440    17
2 D02      70462    23
3 D03      70460    25
4 D04      20956    23
5 D05      20940    27
6 D06      20940    19
7 D07      20955    22
8 D08      20940    18
9 D09      20955    34
10 D10      20940    21
11 D11     170432    33
```

From this quick tibble summary we can see that D04 - D10 (All are around 20,000) are Household based, where D01 - D03 is person based (Around 70,000) and then we have D11 with around 170,000 which is also likely to do with people

## Exploring the data:

```
#View(D01)
head(D01,5)
```

```
# A tibble: 5 x 17
  survey_year province household_id person_number    psu province_code sex_code
    <dbl> <chr>    <chr>    <chr>    <dbl>    <dbl>    <dbl>
1      2024 Western~ 16010009000~ 01      1.60e10      1      1
2      2024 Western~ 16010009000~ 02      1.60e10      1      2
3      2024 Western~ 16010009000~ 01      1.60e10      1      1
4      2024 Western~ 16010009000~ 01      1.60e10      1      1
5      2024 Western~ 16010009000~ 02      1.60e10      1      2
# i 10 more variables: age <dbl>, age_group_code <dbl>,
# population_group_code <dbl>, language_code <dbl>,
# relationship_to_head_code <dbl>, marital_status_code <dbl>,
# geo_type_code <dbl>, metro_code <dbl>, metro_indicator <dbl>,
# person_weight <dbl>
```

```
skim(D01)
```

Table 1: Data summary

|                        |       |
|------------------------|-------|
| Name                   | D01   |
| Number of rows         | 70440 |
| Number of columns      | 17    |
| Column type frequency: |       |
| character              | 3     |
| numeric                | 14    |
| Group variables        | None  |

#### Variable type: character

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---------------|-----------|---------------|-----|-----|-------|----------|------------|
| province      | 0         | 1             | 7   | 13  | 0     | 24       | 0          |
| household_id  | 25        | 1             | 18  | 18  | 0     | 20940    | 0          |
| person_number | 0         | 1             | 2   | 2   | 0     | 25       | 0          |

# Variable type: numeric

| skim_vari-<br>able        | n_miss-<br>ing | com-<br>plete_rat | mean         | sd           | p0          | p25          | p50          | p75          | p100         | hist |
|---------------------------|----------------|-------------------|--------------|--------------|-------------|--------------|--------------|--------------|--------------|------|
| survey_year               | 0              | 1                 | 2.024000e+03 | 0.00000e+00  | 2024        | 2.024000e+03 | 2.024000e+03 | 2.024000e+03 | 2.024000e+03 |      |
| psu                       | 0              | 1                 | 6.154146e+10 | 2.805146e+10 | 0.00000e+00 | 3.080100e+10 | 5.991543e+10 | 9.990479e+10 | 8.70069e+10  |      |
| province_code             | 0              | 1                 | 5.350000e+01 | 0.00000e+00  |             | 3.000000e+01 | 5.000000e+01 | 7.000000e+01 | 9.00000e+01  |      |
| sex_code                  | 0              | 1                 | 1.520000e-01 | 0.00000e-01  | 1           | 1.000000e-01 | 2.000000e-01 | 2.000000e-01 | 2.00000e+00  |      |
| age                       | 90             | 1                 | 3.050000e+02 | 0.95000e+02  |             | 1.300000e+02 | 2.800000e+02 | 4.500000e+02 | 7.50000e+02  |      |
| age_group_code            | 0              | 1                 | 6.680000e+01 | 1.00000e+01  |             | 3.000000e+01 | 6.000000e+01 | 1.000000e+02 | 1.60000e+02  |      |
| population_group_code     | 0              | 1                 | 1.260000e-01 | 0.00000e-01  | 1           | 1.000000e-01 | 1.000000e-01 | 1.000000e-01 | 1.00000e+00  |      |
| language_code             | 0              | 1                 | 5.960000e+01 | 3.32000e+01  |             | 4.000000e+01 | 5.000000e+01 | 8.000000e+01 | 1.40000e+02  |      |
| relationship_to_head_code | 0              | 1                 | 3.490000e+01 | 3.48000e+01  |             | 1.000000e+01 | 3.000000e+01 | 7.000000e+01 | 9.00000e+01  |      |
| marital_status_code       | 0              | 1                 | 5.460000e+01 | 2.40000e+01  |             | 3.000000e+01 | 7.000000e+01 | 7.000000e+01 | 9.00000e+01  |      |
| geo_type_code             | 0              | 1                 | 1.420000e-01 | 5.60000e-01  | 1           | 1.000000e-01 | 1.000000e-01 | 2.000000e-01 | 3.00000e+00  |      |
| metro_code                | 0              | 1                 | 1.015000e+01 | 0.00000e+00  |             | 6.000000e+01 | 1.000000e+02 | 1.500000e+02 | 1.70000e+02  |      |
| metro_indicator           | 0              | 1                 | 1.630000e-01 | 0.80000e-01  | 1           | 1.000000e-01 | 2.000000e-01 | 2.000000e-01 | 2.00000e+00  |      |
| person_weight             | 0              | 1                 | 8.969200e+02 | 1.507000e+02 | 0           | 6.076200e+02 | 8.211700e+02 | 1.087830e+03 | 1.533990e+03 |      |

```
head(D02,5)
```

```
# A tibble: 5 x 23
  survey_year province household_id person_number province_code age sex_code
    <dbl> <chr> <chr> <chr> <dbl> <dbl> <dbl>
1      2024 Western C~ 16010009000~ 01      1      81      1
2      2024 Western C~ 16010009000~ 02      1      78      2
3      2024 Western C~ 16010009000~ 01      1      27      1
4      2024 Western C~ 16010009000~ 01      1      36      1
5      2024 Western C~ 16010009000~ 02      1      30      2
# i 16 more variables: education_level_code <dbl>,
# education_attendance_code <chr>, reason_not_attending_code <dbl>,
# institution_type_code <dbl>, public_private_institution_code <dbl>,
# transport_mode_to_education_code <dbl>, travel_time_to_education <dbl>,
```



```
# bursary_code <dbl>, total_education_fees <dbl>, no_fees_school_code <dbl>,
# grade_code <dbl>, school_food_programme_code <dbl>, absence_code <dbl>,
# absence_days <dbl>, geo_type_code <dbl>, person_weight <dbl>
```

```
skim(D02)
```

Table 2: Data summary

|                        |       |
|------------------------|-------|
| Name                   | D02   |
| Number of rows         | 70462 |
| Number of columns      | 23    |
| Column type frequency: |       |
| character              | 4     |
| numeric                | 19    |
| Group variables        | None  |

#### Variable type: character

| skim_variable                  | n_miss-<br>ing | com-<br>plete_rate | min | max | empty | n_unique | whites-<br>pace |
|--------------------------------|----------------|--------------------|-----|-----|-------|----------|-----------------|
| province                       | 0              | 1                  | 7   | 13  | 0     | 22       | 0               |
| household_id                   | 0              | 1                  | 18  | 18  | 0     | 20940    | 0               |
| person_number                  | 0              | 1                  | 2   | 2   | 0     | 25       | 0               |
| education_atten-<br>dance_code | 0              | 1                  | 1   | 12  | 0     | 7        | 0               |

#### Variable type: numeric

| skim_variable             | n_miss-<br>ing | com-<br>plete_rate | mean        | sd    | p0   | p25     | p50     | p75     | p100    | hist |
|---------------------------|----------------|--------------------|-------------|-------|------|---------|---------|---------|---------|------|
| survey_year               | 0              | 1                  | 2024.000.00 |       | 2024 | 2024.00 | 2024.00 | 2024.00 | 2024.00 |      |
| province_code             | 0              | 1                  | 5.35        | 2.54  | 1    | 3.00    | 5.00    | 7.00    | 9.00    |      |
| age                       | 0              | 1                  | 30.49       | 20.90 | 0    | 13.00   | 28.00   | 45.00   | 110.00  |      |
| sex_code                  | 0              | 1                  | 1.52        | 0.50  | 1    | 1.00    | 2.00    | 2.00    | 2.00    |      |
| educa-<br>tion_level_code | 120            | 1                  | 22.02       | 30.39 | 0    | 8.00    | 11.00   | 12.00   | 99.00   |      |

| skim_variable                              | n_miss-<br>ing | com-<br>plete_rate | mean   | sd     | p0 | p25    | p50    | p75     | p100   | hist |
|--------------------------------------------|----------------|--------------------|--------|--------|----|--------|--------|---------|--------|------|
| reason_not_attend-<br>ing_code             | 0              | 1                  | 36.85  | 40.34  | 1  | 5.00   | 7.00   | 88.00   | 88.00  |      |
| institu-<br>tion_type_code                 | 0              | 1                  | 62.94  | 38.98  | 1  | 2.00   | 88.00  | 88.00   | 88.00  |      |
| public_private_in-<br>stitution_code       | 0              | 1                  | 5.98   | 3.15   | 1  | 1.00   | 8.00   | 8.00    | 8.00   |      |
| trans-<br>port_mode_to_ed-<br>ucation_code | 0              | 1                  | 64.36  | 38.10  | 1  | 8.00   | 88.00  | 88.00   | 88.00  |      |
| travel_time_to_ed-<br>ucation              | 0              | 1                  | 6.30   | 2.77   | 1  | 3.00   | 8.00   | 8.00    | 8.00   |      |
| bursary_code                               | 0              | 1                  | 6.22   | 2.78   | 1  | 2.00   | 8.00   | 8.00    | 8.00   |      |
| total_educ-<br>ation_fees                  | 0              | 1                  | 63.12  | 38.83  | 0  | 9.00   | 88.00  | 88.00   | 88.00  |      |
| no_fees_school_code                        | 0              | 1                  | 7.07   | 1.95   | 1  | 8.00   | 8.00   | 8.00    | 8.00   |      |
| grade_code                                 | 0              | 1                  | 66.72  | 35.92  | 0  | 12.00  | 88.00  | 88.00   | 88.00  |      |
| school_food_pro-<br>gramme_code            | 0              | 1                  | 6.22   | 3.00   | 1  | 2.00   | 8.00   | 8.00    | 8.00   |      |
| absence_code                               | 0              | 1                  | 6.48   | 2.59   | 1  | 4.00   | 8.00   | 8.00    | 8.00   |      |
| absence_days                               | 0              | 1                  | 7.94   | 0.62   | 1  | 8.00   | 8.00   | 8.00    | 8.00   |      |
| geo_type_code                              | 0              | 1                  | 1.42   | 0.56   | 1  | 1.00   | 1.00   | 2.00    | 3.00   |      |
| person_weight                              | 0              | 1                  | 896.96 | 453.74 | 50 | 607.62 | 821.18 | 1087.83 | 530.99 |      |

```
head(D03,5)
```

```
# A tibble: 5 x 25
  survey_year province household_id person_number province_code age sex_code
    <dbl> <chr>      <chr>      <chr>          <dbl> <dbl>   <dbl>
1    2024 Western C~ 16010009000~ 01              1    81      1
2    2024 Western C~ 16010009000~ 02              1    78      2
3    2024 Western C~ 16010009000~ 01              1    27      1
4    2024 Western C~ 16010009000~ 01              1    36      1
5    2024 Western C~ 16010009000~ 02              1    30      2
# i 18 more variables: education_level_code <dbl>, wage_work_code <dbl>,
# business_work_code <dbl>, volunteer_work_code <dbl>, retired_code <dbl>,
# salary_amount_raw <dbl>, salary_period_code <dbl>,
# salary_category_code <dbl>, transport_to_work_code <dbl>,
# minutes_to_work <chr>, sector_code <dbl>, looking_for_work_code <dbl>,
# could_start_work_code <dbl>, employment_status1code <dbl>,
```

```
# employment_status2code <dbl>, lab_salary_raw <dbl>, ...
```

```
skim(D03)
```

Table 3: Data summary

|                        |       |
|------------------------|-------|
| Name                   | D03   |
| Number of rows         | 70460 |
| Number of columns      | 25    |
| Column type frequency: |       |
| character              | 4     |
| numeric                | 21    |
| Group variables        | None  |

#### Variable type: character

| skim_variable   | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|-----------------|-----------|---------------|-----|-----|-------|----------|------------|
| province        | 0         | 1             | 7   | 13  | 0     | 9        | 0          |
| household_id    | 0         | 1             | 18  | 18  | 0     | 20940    | 0          |
| person_number   | 0         | 1             | 2   | 2   | 0     | 25       | 0          |
| minutes_to_work | 0         | 1             | 1   | 12  | 0     | 10       | 0          |

#### Variable type: numeric

| skim_variable        | n_missing | complete_rate | mean    | sd    | p0   | p25    | p50     | p75     | p100        | hist |
|----------------------|-----------|---------------|---------|-------|------|--------|---------|---------|-------------|------|
| survey_year          | 0         | 1             | 2024.00 | 0.00  | 2024 | 2024.0 | 2024.00 | 2024.00 | 2.02400e+03 |      |
| province_code        | 0         | 1             | 5.35    | 2.54  | 1    | 3.0    | 5.00    | 7.00    | 9.00000e+00 |      |
| age                  | 0         | 1             | 30.49   | 20.90 | 0    | 13.0   | 28.00   | 45.00   | 1.10000e+02 |      |
| sex_code             | 0         | 1             | 1.52    | 0.50  | 1    | 1.0    | 2.00    | 2.00    | 2.00000e+00 |      |
| education_level_code | 0         | 1             | 22.02   | 30.39 | 0    | 8.0    | 11.00   | 12.00   | 9.90000e+01 |      |
| wage_work_code       | 0         | 1             | 3.46    | 2.87  | 1    | 2.0    | 2.00    | 8.00    | 9.00000e+00 |      |
| business_work_code   | 0         | 1             | 3.64    | 2.73  | 1    | 2.0    | 2.00    | 8.00    | 9.00000e+00 |      |

[illegible]

```
# A tibble: 5 x 23
  survey_year province household_id province_code head_population_grou~1
      <dbl> <chr>      <chr>          <dbl>          <dbl>
1      2024 Western Cape 1601000900000003~          1          4
2      2024 Western Cape 1601000900000007~          1          1
3      2024 Western Cape 1601000900000013~          1          1
4      2024 Western Cape 1601003300000001~          1          2
5      2024 Western Cape 1601003300000006~          1          2
# i abbreviated name: 1: head_population_group_code
# i 18 more variables: head_sex_code <dbl>, head_age <dbl>,
#   main_dwelling_code <dbl>, wall_material_code <dbl>,
```

```
# roof_material_code <dbl>, floor_material_code <dbl>,
# wall_condition_code <dbl>, roof_condition_code <dbl>,
# floor_condition_code <dbl>, total_rooms <dbl>, tenure_code <dbl>,
# rent_raw <dbl>, rdp_housing_code <dbl>, housing_subsidy_code <dbl>, ...
```

```
skim(D04)
```

Table 4: Data summary

|                        |       |
|------------------------|-------|
| Name                   | D04   |
| Number of rows         | 20956 |
| Number of columns      | 23    |
| Column type frequency: |       |
| character              | 2     |
| numeric                | 21    |
| Group variables        | None  |

#### Variable type: character

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---------------|-----------|---------------|-----|-----|-------|----------|------------|
| province      | 0         | 1             | 7   | 13  | 0     | 23       | 0          |
| household_id  | 0         | 1             | 18  | 18  | 0     | 20940    | 0          |

#### Variable type: numeric

| skim_variable              | n_missing | complete_rate | mean    | sd    | p0      | p25     | p50     | p75    | p100    | hist |
|----------------------------|-----------|---------------|---------|-------|---------|---------|---------|--------|---------|------|
| survey_year                | 0         | 1             | 2024.00 | 0.00  | 2024.00 | 2024.00 | 2024.00 | 2024.0 | 2024.00 |      |
| province_code              | 0         | 1             | 5.39    | 2.52  | 1.00    | 3.00    | 6.00    | 7.0    | 9.00    |      |
| head_population_group_code | 0         | 1             | 1.30    | 0.78  | 1.00    | 1.00    | 1.00    | 1.0    | 4.00    |      |
| head_sex_code              | 0         | 1             | 1.44    | 0.50  | 1.00    | 1.00    | 1.00    | 2.0    | 2.00    |      |
| head_age                   | 0         | 1             | 50.09   | 15.64 | 14.00   | 38.00   | 49.00   | 62.0   | 107.00  |      |
| main_dwelling_code         | 0         | 1             | 2.73    | 3.03  | 1.00    | 1.00    | 1.00    | 3.0    | 12.00   |      |
| wall_material_code         | 55        | 1             | 1.85    | 1.84  | 1.00    | 1.00    | 1.00    | 2.0    | 99.00   |      |

| skim_variable        | n_missing | complete_rate | mean   | sd     | p0     | p25    | p50    | p75    | p100    | hist |
|----------------------|-----------|---------------|--------|--------|--------|--------|--------|--------|---------|------|
| roof_material_code   | 0         | 1             | 4.14   | 4.04   | 1.00   | 2.00   | 2.00   | 8.0    | 99.00   |      |
| floor_material_code  | 0         | 1             | 6.22   | 1.12   | 1.00   | 6.00   | 6.00   | 7.0    | 9.00    |      |
| wall_condition_code  | 0         | 1             | 3.69   | 0.86   | 1.00   | 3.00   | 4.00   | 4.0    | 5.00    |      |
| roof_condition_code  | 0         | 1             | 3.64   | 0.88   | 1.00   | 3.00   | 4.00   | 4.0    | 9.00    |      |
| floor_condition_code | 0         | 1             | 3.74   | 0.83   | 1.00   | 3.00   | 4.00   | 4.0    | 9.00    |      |
| total_rooms          | 0         | 1             | 4.99   | 3.16   | 1.00   | 3.00   | 5.00   | 7.0    | 99.00   |      |
| tenure_code          | 0         | 1             | 4.28   | 1.83   | 1.00   | 5.00   | 5.00   | 5.0    | 99.00   |      |
| rent_raw             | 0         | 1             | 67.07  | 36.51  | 1.00   | 88.00  | 88.00  | 88.0   | 99.00   |      |
| rdp_housing_code     | 0         | 1             | 1.81   | 0.40   | 1.00   | 2.00   | 2.00   | 2.0    | 9.00    |      |
| housing_subsidy_code | 0         | 1             | 3.06   | 2.45   | 1.00   | 2.00   | 2.00   | 2.0    | 9.00    |      |
| household_size       | 0         | 1             | 3.36   | 2.33   | 0.00   | 2.00   | 3.00   | 5.0    | 25.00   |      |
| geo_type_code        | 0         | 1             | 1.38   | 0.55   | 1.00   | 1.00   | 1.00   | 2.0    | 3.00    |      |
| metro_code           | 0         | 1             | 10.25  | 5.04   | 1.00   | 6.00   | 11.00  | 14.0   | 17.00   |      |
| household_weight     | 0         | 1             | 933.72 | 426.48 | 191.85 | 625.38 | 874.59 | 1117.7 | 5049.94 |      |

```
head(D05,5)
```

```
# A tibble: 5 x 27
  survey_year province household_id province_code drinking_water_source_code
  <dbl> <chr> <chr> <dbl> <dbl>
1 2024 Western Cape 1601000900000003~ 1 1
2 2024 Western Cape 1601000900000007~ 1 1
3 2024 Western Cape 1601000900000013~ 1 2
4 2024 Western Cape 1601003300000001~ 1 1
5 2024 Western Cape 1601003300000006~ 1 1
# i abbreviated name: 1: drinking_water_source_code
# i 22 more variables: distance_to_water_code <dbl>, time_to_water_raw <chr>,
# piped_water_code <dbl>, water_safe_code <dbl>, water_clear_code <dbl>,
# water_taste_code <dbl>, water_smell_code <dbl>, water_treatment_code <dbl>,
# municipal_water_source_code <dbl>, water_meter_code <dbl>,
# pays_for_water_code <dbl>, water_interruptions12months_code <dbl>,
```

```
# water_interruption_frequency_code <dbl>, ...
```

```
skim(D05)
```

Table 5: Data summary

|                        |       |
|------------------------|-------|
| Name                   | D05   |
| Number of rows         | 20940 |
| Number of columns      | 27    |
| Column type frequency: |       |
| character              | 3     |
| numeric                | 24    |
| Group variables        | None  |

#### Variable type: character

| skim_variable          | n_miss-<br>ing | com-<br>plete_rate | min | max | empty | n_unique | whitespace |
|------------------------|----------------|--------------------|-----|-----|-------|----------|------------|
| province               | 0              | 1                  | 7   | 13  | 0     | 19       | 0          |
| household_id           | 0              | 1                  | 16  | 18  | 0     | 20940    | 0          |
| time_to_wa-<br>ter_raw | 0              | 1                  | 1   | 12  | 0     | 11       | 0          |

#### Variable type: numeric

| skim_variable                   | n_miss-<br>ing | com-<br>plete_rate | mean     | sd   | p0      | p25     | p50     | p75    | p100    | hist |
|---------------------------------|----------------|--------------------|----------|------|---------|---------|---------|--------|---------|------|
| survey_year                     | 0              | 1                  | 2024.000 | 0.00 | 2024.00 | 2024.00 | 2024.00 | 2024.0 | 2024.00 |      |
| province_code                   | 0              | 1                  | 5.39     | 2.52 | 1.00    | 3.00    | 6.00    | 7.0    | 9.00    |      |
| drinking_wa-<br>ter_source_code | 60             | 1                  | 2.79     | 2.78 | 1.00    | 1.00    | 2.00    | 3.0    | 16.00   |      |
| distance_to_wa-<br>ter_code     | 0              | 1                  | 6.73     | 2.58 | 1.00    | 8.00    | 8.00    | 8.0    | 9.00    |      |
| pipeds_water_code               | 0              | 1                  | 6.73     | 2.55 | 1.00    | 8.00    | 8.00    | 8.0    | 9.00    |      |
| water_safe_code                 | 0              | 1                  | 1.16     | 0.37 | 1.00    | 1.00    | 1.00    | 1.0    | 9.00    |      |
| water_clear_code                | 0              | 1                  | 1.16     | 0.38 | 1.00    | 1.00    | 1.00    | 1.0    | 9.00    |      |
| water_taste_code                | 0              | 1                  | 1.18     | 0.39 | 1.00    | 1.00    | 1.00    | 1.0    | 9.00    |      |

| skim_variable                               | n_miss-<br>ing | com-<br>plete_rate | mean   | sd     | p0     | p25    | p50    | p75     | p100    | hist |
|---------------------------------------------|----------------|--------------------|--------|--------|--------|--------|--------|---------|---------|------|
| water_smell_code                            | 0              | 1                  | 1.16   | 0.37   | 1.00   | 1.00   | 1.00   | 1.0     | 9.00    |      |
| water_treat-<br>ment_code                   | 0              | 1                  | 2.76   | 0.58   | 1.00   | 3.00   | 3.00   | 3.0     | 9.00    |      |
| municipal_wa-<br>ter_source_code            | 0              | 1                  | 1.24   | 0.45   | 1.00   | 1.00   | 1.00   | 1.0     | 9.00    |      |
| water_meter_code                            | 0              | 1                  | 2.95   | 2.80   | 1.00   | 1.00   | 2.00   | 2.0     | 8.00    |      |
| pays_for_wa-<br>ter_code                    | 0              | 1                  | 3.06   | 2.74   | 1.00   | 1.00   | 2.00   | 2.0     | 8.00    |      |
| water_interrup-<br>tions12months_code       | 0              | 1                  | 2.94   | 2.81   | 1.00   | 1.00   | 2.00   | 2.0     | 8.00    |      |
| water_interrup-<br>tion_fre-<br>quency_code | 0              | 1                  | 5.60   | 2.82   | 1.00   | 2.00   | 8.00   | 8.0     | 8.00    |      |
| water_alterna-<br>tive_source_code          | 0              | 1                  | 66.93  | 34.96  | 1.00   | 11.00  | 88.00  | 88.0    | 88.00   |      |
| main_toilet_code                            | 0              | 1                  | 2.85   | 2.58   | 1.00   | 1.00   | 1.00   | 5.0     | 99.00   |      |
| shared_toilet_code                          | 0              | 1                  | 1.82   | 0.69   | 1.00   | 2.00   | 2.00   | 2.0     | 9.00    |      |
| toilet_loca-<br>tion_code                   | 0              | 1                  | 1.66   | 0.79   | 1.00   | 1.00   | 2.00   | 2.0     | 9.00    |      |
| toilet_dis-<br>tance_code                   | 0              | 1                  | 7.79   | 1.17   | 1.00   | 8.00   | 8.00   | 8.0     | 8.00    |      |
| handwashing_facil-<br>ity_code              | 0              | 1                  | 1.40   | 0.76   | 1.00   | 1.00   | 1.00   | 2.0     | 9.00    |      |
| handwashing_code                            | 0              | 1                  | 1.65   | 0.82   | 1.00   | 1.00   | 2.00   | 2.0     | 9.00    |      |
| geo_type_code                               | 0              | 1                  | 1.38   | 0.55   | 1.00   | 1.00   | 1.00   | 2.0     | 3.00    |      |
| household_weight                            | 0              | 1                  | 933.68 | 426.29 | 191.85 | 625.38 | 874.59 | 1117.75 | 5049.94 |      |

```
head(D06,5)
```

```
# A tibble: 5 x 19
```

|   | survey_year | province     | household_id      | province_code | electricity_access_c~1 |
|---|-------------|--------------|-------------------|---------------|------------------------|
|   | <dbl>       | <chr>        | <chr>             | <dbl>         | <dbl>                  |
| 1 | 2024        | Western Cape | 1601000900000003~ | 1             | 1                      |
| 2 | 2024        | Western Cape | 1601000900000007~ | 1             | 1                      |
| 3 | 2024        | Western Cape | 1601000900000013~ | 1             | 1                      |
| 4 | 2024        | Western Cape | 1601003300000001~ | 1             | 1                      |
| 5 | 2024        | Western Cape | 1601003300000006~ | 1             | 1                      |

```
# i abbreviated name: 1: electricity_access_code
```

```
# i 14 more variables: main_electricity_source_code <dbl>,
```



```
# mains_electricity_code <dbl>, electricity_supply_code <dbl>,
# cooking_energy_code <chr>, lighting_energy_code <dbl>,
# space_heating_energy_code <dbl>, loadshedding_frequency_code <dbl>,
# alternative_lighting_code <dbl>, alternative_cooking_code <dbl>,
# pays_for_electricity_code <dbl>, solar_geyser_asset_code <dbl>, ...
```

```
skim(D06)
```

Table 6: Data summary

|                        |       |
|------------------------|-------|
| Name                   | D06   |
| Number of rows         | 20940 |
| Number of columns      | 19    |
| Column type frequency: |       |
| character              | 3     |
| numeric                | 16    |
| Group variables        | None  |

#### Variable type: character

| skim_variable            | n_miss-<br>ing | com-<br>plete_rate | min | max | empty | n_unique | whitespace |
|--------------------------|----------------|--------------------|-----|-----|-------|----------|------------|
| province                 | 0              | 1                  | 7   | 13  | 0     | 22       | 0          |
| household_id             | 0              | 1                  | 18  | 18  | 0     | 20940    | 0          |
| cooking_en-<br>ergy_code | 0              | 1                  | 1   | 12  | 0     | 12       | 0          |

#### Variable type: numeric

| skim_variable                     | n_miss-<br>ing | com-<br>plete_rate | mean    | sd   | p0      | p25     | p50     | p75    | p100    | hist |
|-----------------------------------|----------------|--------------------|---------|------|---------|---------|---------|--------|---------|------|
| survey_year                       | 0              | 1                  | 2024.00 | 0.00 | 2024.00 | 2024.00 | 2024.00 | 2024.0 | 2024.00 |      |
| province_code                     | 0              | 1                  | 5.39    | 2.52 | 1.00    | 3.00    | 6.00    | 7.0    | 9.00    |      |
| electricity_ac-<br>cess_code      | 50             | 1                  | 1.05    | 0.23 | 1.00    | 1.00    | 1.00    | 1.0    | 9.00    |      |
| main_electric-<br>ity_source_code | 0              | 1                  | 2.32    | 1.43 | 1.00    | 2.00    | 2.00    | 2.0    | 8.00    |      |

| skim_variable               | n_miss-<br>ing | com-<br>plete_rate | mean   | sd     | p0     | p25    | p50    | p75    | p100    | hist |
|-----------------------------|----------------|--------------------|--------|--------|--------|--------|--------|--------|---------|------|
| mains_electricity_code      | 0              | 1                  | 1.42   | 1.61   | 1.00   | 1.00   | 1.00   | 1.0    | 8.00    |      |
| electricity_supply_code     | 0              | 1                  | 2.16   | 1.88   | 1.00   | 1.00   | 2.00   | 2.0    | 8.00    |      |
| lighting_energy_code        | 0              | 1                  | 1.34   | 1.35   | 1.00   | 1.00   | 1.00   | 1.0    | 10.00   |      |
| space_heating_energy_code   | 0              | 1                  | 4.05   | 3.44   | 1.00   | 1.00   | 2.00   | 9.0    | 10.00   |      |
| loadshedding_frequency_code | 0              | 1                  | 3.85   | 1.66   | 1.00   | 2.00   | 5.00   | 5.0    | 9.00    |      |
| alternative_lighting_code   | 0              | 1                  | 57.48  | 39.80  | 1.00   | 8.00   | 88.00  | 88.0   | 88.00   |      |
| alternative_cooking_code    | 0              | 1                  | 56.93  | 40.50  | 1.00   | 4.00   | 88.00  | 88.0   | 88.00   |      |
| pays_for_electricity_code   | 0              | 1                  | 1.73   | 1.97   | 1.00   | 1.00   | 1.00   | 1.0    | 8.00    |      |
| solar_geyser_asset_code     | 0              | 1                  | 1.97   | 0.19   | 1.00   | 2.00   | 2.00   | 2.0    | 9.00    |      |
| solar_panel_asset_code      | 0              | 1                  | 1.98   | 0.17   | 1.00   | 2.00   | 2.00   | 2.0    | 9.00    |      |
| geo_type_code               | 0              | 1                  | 1.38   | 0.55   | 1.00   | 1.00   | 1.00   | 2.0    | 3.00    |      |
| household_weight            | 0              | 1                  | 933.68 | 426.29 | 191.85 | 625.38 | 874.59 | 1117.7 | 5049.94 |      |

```
head(D07,5)
```

```
# A tibble: 5 x 22
  survey_year province household_id province_code adult_hunger_code
    <dbl> <chr>      <chr>          <dbl>          <dbl>
1    2024 Western Cape 160100090000003901          1          1
2    2024 Western Cape 160100090000007201          1          1
3    2024 Western Cape 160100090000013801          1          1
4    2024 Western Cape 160100330000001101          1          1
5    2024 Western Cape 160100330000006101          1          1
# i 17 more variables: child_hunger_code <dbl>,
#   worried_food_would_run_out_code <dbl>,
#   unable_to_eat_healthy_food_code <dbl>, ate_few_foods_code <dbl>,
#   few_foods_days <dbl>, skipped_meal_code <dbl>, skipped_meal_days <dbl>,
#   ate_less_code <dbl>, ate_less_days <dbl>, ran_out_of_food_code <dbl>,
```

```
#   ran_out_days <dbl>, hungry_but_did_not_eat_code <dbl>,
#   whole_day_without_eating_code <dbl>, household_size <dbl>, ...
```

```
skim(D07)
```

Table 7: Data summary

|                        |       |
|------------------------|-------|
| Name                   | D07   |
| Number of rows         | 20955 |
| Number of columns      | 22    |
| Column type frequency: |       |
| character              | 2     |
| numeric                | 20    |
| Group variables        | None  |

#### Variable type: character

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---------------|-----------|---------------|-----|-----|-------|----------|------------|
| province      | 0         | 1             | 7   | 13  | 0     | 9        | 0          |
| household_id  | 0         | 1             | 18  | 18  | 0     | 20940    | 0          |

#### Variable type: numeric

| skim_variable                   | n_missing | complete_rate | mean    | sd   | p0      | p25     | p50     | p75     | p100        | hist        |
|---------------------------------|-----------|---------------|---------|------|---------|---------|---------|---------|-------------|-------------|
| survey_year                     | 0         | 1             | 2024.00 | 0.00 | 2024.00 | 2024.00 | 2024.00 | 2024.00 | 2024.00     | 2.02400e+03 |
| province_code                   | 0         | 1             | 5.39    | 2.52 | 1.00    | 3.00    | 6.00    | 7.0     | 9.00000e+00 |             |
| adult_hunger_code               | 65        | 1             | 1.38    | 0.88 | 1.00    | 1.00    | 1.00    | 1.0     | 9.00000e+00 |             |
| child_hunger_code               | 0         | 1             | 3.34    | 2.38 | 1.00    | 1.00    | 3.00    | 6.0     | 9.00000e+00 |             |
| worried_food_would_run_out_code | 0         | 1             | 1.76    | 0.51 | 1.00    | 2.00    | 2.00    | 2.0     | 9.00000e+00 |             |
| unable_to_eat_healthy_food_code | 0         | 1             | 1.76    | 0.51 | 1.00    | 2.00    | 2.00    | 2.0     | 9.00000e+00 |             |
| ate_few_foods_code              | 0         | 1             | 1.76    | 0.48 | 1.00    | 2.00    | 2.00    | 2.0     | 9.00000e+00 |             |
| few_foods_days                  | 0         | 1             | 6.41    | 2.80 | 1.00    | 8.00    | 8.00    | 8.0     | 8.00000e+00 |             |
| skipped_meal_code               | 0         | 1             | 1.84    | 0.45 | 1.00    | 2.00    | 2.00    | 2.0     | 9.00000e+00 |             |
| skipped_meal_days               | 0         | 1             | 6.89    | 2.46 | 1.00    | 8.00    | 8.00    | 8.0     | 8.00000e+00 |             |

| skim_variable            | n_missing | complete_rate | mean       | sd         | p0     | p25     | p50     | p75     | p100        | hist |
|--------------------------|-----------|---------------|------------|------------|--------|---------|---------|---------|-------------|------|
| ate_less_code            | 0         | 1             | 1.81       | 0.48       | 1.00   | 2.00    | 2.00    | 2.0     | 9.00000e+00 |      |
| ate_less_days            | 0         | 1             | 6.71       | 2.60       | 1.00   | 8.00    | 8.00    | 8.0     | 8.00000e+00 |      |
| ran_out_of_food_code     | 0         | 1             | 1.83       | 0.47       | 1.00   | 2.00    | 2.00    | 2.0     | 9.00000e+00 |      |
| ran_out_days             | 0         | 1             | 6.81       | 2.51       | 1.00   | 8.00    | 8.00    | 8.0     | 8.00000e+00 |      |
| hun-                     | 0         | 1             | 1.88       | 0.43       | 1.00   | 2.00    | 2.00    | 2.0     | 9.00000e+00 |      |
| gry_but_did_not_eat_code |           |               |            |            |        |         |         |         |             |      |
| whole_day_with-          | 0         | 1             | 1.92       | 0.40       | 1.00   | 2.00    | 2.00    | 2.0     | 9.00000e+00 |      |
| out_eating_code          |           |               |            |            |        |         |         |         |             |      |
| household_size           | 0         | 1             | 3.36       | 2.33       | 0.00   | 2.00    | 3.00    | 5.0     | 2.50000e+01 |      |
| to-                      | 0         | 1             | 1163011.72 | 1554100300 | 0.00   | 2470.00 | 5257.00 | 12292.5 | 1.00000e+10 |      |
| tal_monthly_house-       |           |               |            |            |        |         |         |         |             |      |
| hold_income_raw          |           |               |            |            |        |         |         |         |             |      |
| geo_type_code            | 0         | 1             | 1.38       | 0.55       | 1.00   | 1.00    | 1.00    | 2.0     | 3.00000e+00 |      |
| household_weight         | 0         | 1             | 933.61     | 426.23     | 191.85 | 625.38  | 874.59  | 1117.7  | 5.04994e+03 |      |

```
head(D08,5)
```

```
# A tibble: 5 x 18
  survey_year province household_id province_code telephone_code cellphone_code
    <dbl> <chr>      <chr>          <dbl>          <dbl>          <dbl>
1    2024 Western ~ 16010009000~          1            1            1
2    2024 Western ~ 16010009000~          1            2            1
3    2024 Western ~ 16010009000~          1            2            1
4    2024 Western ~ 16010033000~          1            2            1
5    2024 Western ~ 16010033000~          1            2            1
# i 12 more variables: fixed_internet_code <dbl>, mobile_internet_code <dbl>,
#   public_wi-fi_internet_code <dbl>, study_internet_code <dbl>,
#   work_internet_code <dbl>, library_internet_code <dbl>,
#   cafe_internet_code <dbl>, household_mobile_phones <dbl>,
#   household_smartphones <dbl>, computer_asset_code <dbl>,
#   geo_type_code <dbl>, household_weight <dbl>
```

```
skim(D08)
```

Table 8: Data summary

|                |       |
|----------------|-------|
| Name           | D08   |
| Number of rows | 20940 |

|                        |      |
|------------------------|------|
| Number of columns      | 18   |
| <hr/>                  |      |
| Column type frequency: |      |
| character              | 2    |
| numeric                | 16   |
| <hr/>                  |      |
| Group variables        | None |
| <hr/>                  |      |

### Variable type: character

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---------------|-----------|---------------|-----|-----|-------|----------|------------|
| province      | 0         | 1             | 7   | 13  | 0     | 24       | 0          |
| household_id  | 0         | 1             | 18  | 18  | 0     | 20940    | 0          |

### Variable type: numeric

| skim_variable               | n_missing | complete_rate | mean    | sd   | p0      | p25     | p50     | p75    | p100    | hist |
|-----------------------------|-----------|---------------|---------|------|---------|---------|---------|--------|---------|------|
| survey_year                 | 0         | 1             | 2024.00 | 0.00 | 2024.00 | 2024.00 | 2024.00 | 2024.0 | 2024.00 |      |
| province_code               | 0         | 1             | 5.39    | 2.52 | 1.00    | 3.00    | 6.00    | 7.0    | 9.00    |      |
| telephone_code              | 0         | 1             | 1.97    | 0.18 | 1.00    | 2.00    | 2.00    | 2.0    | 9.00    |      |
| cellphone_code              | 0         | 1             | 1.04    | 0.20 | 1.00    | 1.00    | 1.00    | 1.0    | 2.00    |      |
| fixed_inter-net_code        | 0         | 1             | 1.86    | 0.35 | 1.00    | 2.00    | 2.00    | 2.0    | 9.00    |      |
| mobile_inter-net_code       | 60        | 1             | 1.26    | 0.45 | 1.00    | 1.00    | 1.00    | 2.0    | 9.00    |      |
| public_wi-fi_inter-net_code | 0         | 1             | 1.94    | 0.25 | 1.00    | 2.00    | 2.00    | 2.0    | 9.00    |      |
| study_inter-net_code        | 0         | 1             | 1.97    | 0.19 | 1.00    | 2.00    | 2.00    | 2.0    | 9.00    |      |
| work_inter-net_code         | 0         | 1             | 1.90    | 0.31 | 1.00    | 2.00    | 2.00    | 2.0    | 9.00    |      |
| library_inter-net_code      | 0         | 1             | 1.98    | 0.15 | 1.00    | 2.00    | 2.00    | 2.0    | 9.00    |      |
| cafe_inter-net_code         | 0         | 1             | 1.95    | 0.23 | 1.00    | 2.00    | 2.00    | 2.0    | 9.00    |      |
| household_mobile_phones     | 0         | 1             | 2.15    | 1.39 | 0.00    | 1.00    | 2.00    | 3.0    | 15.00   |      |

| skim_variable                   | n_miss-<br>ing | com-<br>plete_rate | mean   | sd     | p0         | p25    | p50    | p75    | p100    | hist |
|---------------------------------|----------------|--------------------|--------|--------|------------|--------|--------|--------|---------|------|
| house-<br>hold_smart-<br>phones | 0              | 1                  | 1.76   | 1.42   | -<br>20.00 | 1.00   | 2.00   | 2.0    | 13.00   |      |
| computer_as-<br>set_code        | 0              | 1                  | 1.79   | 0.42   | 1.00       | 2.00   | 2.00   | 2.0    | 9.00    |      |
| geo_type_code                   | 0              | 1                  | 1.38   | 0.55   | 1.00       | 1.00   | 1.00   | 2.0    | 3.00    |      |
| house-<br>hold_weight           | 0              | 1                  | 933.68 | 426.29 | 191.85     | 625.38 | 874.59 | 1117.7 | 5049.94 |      |

```
head(D09,5)
```

```
# A tibble: 5 x 34
  survey_year province household_id province_code income_salary_code
    <dbl> <chr>      <chr>          <dbl>          <dbl>
1    2024 Western Cape 160100090000003901          1            2
2    2024 Western Cape 160100090000007201          1            1
3    2024 Western Cape 160100090000013801          1            1
4    2024 Western Cape 160100330000001101          1            1
5    2024 Western Cape 160100330000006101          1            2
# i 29 more variables: income_business_code <dbl>,
# income_remittance_code <dbl>, income_pension_code <dbl>,
# income_grant_code <dbl>, income_agriculture_code <dbl>,
# no_income_code <dbl>, other_income_code <dbl>,
# main_income_source_code <dbl>, required_income_raw <dbl>,
# comparative_income_code <dbl>, expenditure_band_code <dbl>,
# total_monthly_household_income_raw <dbl>, vehicle_asset_code <dbl>, ...
```

```
skim(D09)
```

Table 9: Data summary

|                        |       |
|------------------------|-------|
| Name                   | D09   |
| Number of rows         | 20955 |
| Number of columns      | 34    |
| Column type frequency: |       |
| character              | 3     |
| numeric                | 31    |

|                 |      |
|-----------------|------|
| Group variables | None |
|-----------------|------|

**Variable type: character**

| skim_variable  | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|----------------|-----------|---------------|-----|-----|-------|----------|------------|
| province       | 0         | 1             | 7   | 13  | 0     | 9        | 0          |
| household_id   | 0         | 1             | 18  | 18  | 0     | 20940    | 0          |
| total_vehicles | 0         | 1             | 1   | 12  | 0     | 15       | 0          |

**Variable type: numeric**

| skim_variable           | n_missing | complete_rate | mean       | sd        | p0      | p25     | p50     | p75      | p100         | hist         |
|-------------------------|-----------|---------------|------------|-----------|---------|---------|---------|----------|--------------|--------------|
| survey_year             | 0         | 1             | 2024.00    | 0.00      | 2024.00 | 2024.00 | 2024.00 | 2024.00  | 2024.00      | 2.024000e+03 |
| province_code           | 0         | 1             | 5.39       | 2.52      | 1.00    | 3.00    | 6.00    | 7.0      | 9.000000e+00 |              |
| income_salary_code      | 0         | 1             | 1.42       | 0.49      | 1.00    | 1.00    | 1.00    | 2.0      | 2.000000e+00 |              |
| income_business_code    | 0         | 1             | 1.85       | 0.35      | 1.00    | 2.00    | 2.00    | 2.0      | 2.000000e+00 |              |
| income_remittance_code  | 0         | 1             | 1.87       | 0.34      | 1.00    | 2.00    | 2.00    | 2.0      | 2.000000e+00 |              |
| income_pension_code     | 0         | 1             | 1.96       | 0.20      | 1.00    | 2.00    | 2.00    | 2.0      | 2.000000e+00 |              |
| income_grant_code       | 0         | 1             | 1.41       | 0.49      | 1.00    | 1.00    | 1.00    | 2.0      | 2.000000e+00 |              |
| income_agriculture_code | 0         | 1             | 2.00       | 0.05      | 1.00    | 2.00    | 2.00    | 2.0      | 2.000000e+00 |              |
| no_income_code          | 0         | 1             | 1.99       | 0.11      | 1.00    | 2.00    | 2.00    | 2.0      | 9.000000e+00 |              |
| other_income_code       | 0         | 1             | 1.98       | 0.15      | 1.00    | 2.00    | 2.00    | 2.0      | 2.000000e+00 |              |
| main_income_source_code | 65        | 1             | 2.72       | 2.33      | 1.00    | 1.00    | 2.00    | 5.0      | 9.900000e+01 |              |
| required_income_raw     | 0         | 1             | 3354285.17 | 318039.90 | 1000.00 | 3000.00 | 6000.00 | 15000.00 | 1.000000e+10 |              |
| comparative_income_code | 0         | 1             | 3.38       | 1.01      | 1.00    | 3.00    | 3.00    | 4.0      | 9.000000e+00 |              |
| expenditure_band_code   | 0         | 1             | 8.05       | 2.26      | 1.00    | 7.00    | 8.00    | 9.0      | 9.900000e+01 |              |

| skim_variable                      | n_missing | complete_rate | mean      | sd        | p0     | p25     | p50     | p75      | p100         | hist |
|------------------------------------|-----------|---------------|-----------|-----------|--------|---------|---------|----------|--------------|------|
| total_monthly_household_income_raw | 0         | 1             | 218156.56 | 421147.72 | 0.00   | 2470.00 | 5257.00 | 12220.00 | 9.999999e+06 |      |
| vehicle_asset_code                 | 0         | 1             | 1.73      | 0.45      | 1.00   | 1.00    | 2.00    | 2.0      | 9.000000e+00 |      |
| tv_asset_code                      | 0         | 1             | 1.22      | 0.43      | 1.00   | 1.00    | 1.00    | 1.0      | 9.000000e+00 |      |
| pay_tv_asset_code                  | 0         | 1             | 1.40      | 0.50      | 1.00   | 1.00    | 1.00    | 2.0      | 9.000000e+00 |      |
| computer_asset_code                | 0         | 1             | 1.78      | 0.42      | 1.00   | 2.00    | 2.00    | 2.0      | 9.000000e+00 |      |
| washing_machine_asset_code         | 0         | 1             | 1.61      | 0.50      | 1.00   | 1.00    | 2.00    | 2.0      | 9.000000e+00 |      |
| freezer_asset_code                 | 0         | 1             | 1.76      | 0.44      | 1.00   | 2.00    | 2.00    | 2.0      | 9.000000e+00 |      |
| fridge_asset_code                  | 0         | 1             | 1.19      | 0.40      | 1.00   | 1.00    | 1.00    | 1.0      | 9.000000e+00 |      |
| microwave_asset_code               | 0         | 1             | 1.42      | 0.50      | 1.00   | 1.00    | 1.00    | 2.0      | 9.000000e+00 |      |
| geyser_asset_code                  | 0         | 1             | 1.76      | 0.43      | 1.00   | 2.00    | 2.00    | 2.0      | 9.000000e+00 |      |
| solar_panel_asset_code             | 0         | 1             | 1.98      | 0.17      | 1.00   | 2.00    | 2.00    | 2.0      | 9.000000e+00 |      |
| radio_asset_code                   | 0         | 1             | 1.68      | 0.48      | 1.00   | 1.00    | 2.00    | 2.0      | 9.000000e+00 |      |
| household_size                     | 0         | 1             | 3.36      | 2.33      | 1.00   | 2.00    | 3.00    | 5.0      | 2.500000e+01 |      |
| salary_earners_in_household        | 0         | 1             | 51427.43  | 46610.27  | 0.00   | 0.00    | 2800.00 | 9186.00  | 9.999999e+06 |      |
| social_grant_recipients            | 0         | 1             | 1.52      | 2.04      | 0.00   | 0.00    | 1.00    | 2.0      | 9.900000e+01 |      |
| total_grant_recipients             | 0         | 1             | 1236.60   | 1574.93   | 0.00   | 0.00    | 480.00  | 1990.00  | 1.298000e+04 |      |
| household_weight                   | 0         | 1             | 933.66    | 426.27    | 191.85 | 625.38  | 874.59  | 1117.75  | 5.049940e+03 |      |

```
head(D10,5)
```

```
# A tibble: 5 x 21
  survey_year province household_id province_code taxi_use_code bus_use_code
  <dbl> <chr> <chr> <dbl> <dbl> <dbl>
1 2024 Western Cape 16010009000~ 1 2 2
```



```

2      2024 Western Cape 16010009000~      1      2      2
3      2024 Western Cape 16010009000~      1      2      2
4      2024 Western Cape 16010033000~      1      2      2
5      2024 Western Cape 16010033000~      1      2      2
# i 15 more variables: train_use_code <dbl>, taxi_trips <dbl>,
#   taxi_cost_raw <dbl>, taxi_distance_raw <dbl>, bus_trips <dbl>,
#   bus_cost_raw <dbl>, bus_distance_raw <dbl>, train_trips <dbl>,
#   train_cost_raw <dbl>, train_distance_raw <dbl>,
#   nearest_health_facility_code <dbl>,
#   transport_to_health_facility_code <dbl>, time_to_health_facility_raw <dbl>,
#   geo_type_code <dbl>, household_weight <dbl>

```

```
skim(D10)
```

Table 10: Data summary

|                        |       |
|------------------------|-------|
| Name                   | D10   |
| Number of rows         | 20940 |
| Number of columns      | 21    |
| Column type frequency: |       |
| character              | 2     |
| numeric                | 19    |
| Group variables        | None  |

#### Variable type: character

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---------------|-----------|---------------|-----|-----|-------|----------|------------|
| province      | 0         | 1             | 7   | 13  | 0     | 22       | 0          |
| household_id  | 0         | 1             | 18  | 18  | 0     | 20940    | 0          |

#### Variable type: numeric

| skim_variable | n_missing | complete_rate | mean    | sd   | p0      | p25     | p50     | p75    | p100    | hist |
|---------------|-----------|---------------|---------|------|---------|---------|---------|--------|---------|------|
| survey_year   | 0         | 1             | 2024.00 | 0.00 | 2024.00 | 2024.00 | 2024.00 | 2024.0 | 2024.00 |      |
| province_code | 0         | 1             | 5.39    | 2.52 | 1.00    | 3.00    | 6.00    | 7.0    | 9.00    |      |
| taxi_use_code | 0         | 1             | 1.65    | 0.48 | 1.00    | 1.00    | 2.00    | 2.0    | 9.00    |      |

| skim_variable                               | n_miss-<br>ing | com-<br>plete_rate | mean           | sd     | p0         | p25     | p50    | p75     | p100   | hist |
|---------------------------------------------|----------------|--------------------|----------------|--------|------------|---------|--------|---------|--------|------|
| bus_use_code                                | 0              | 1                  | 1.96           | 0.21   | 1.00       | 2.00    | 2.00   | 2.0     | 9.00   |      |
| train_use_code                              | 0              | 1                  | 2.00           | 0.09   | 1.00       | 2.00    | 2.00   | 2.0     | 9.00   |      |
| taxi_trips                                  | 0              | 1                  | 582.91         | 418.91 | 1.00       | 10.00   | 888.00 | 888.0   | 888.00 |      |
| taxi_cost_raw                               | 0              | 1                  | 5885.474128.81 | 1.00   | 250.00     | 8888.00 | 8888.0 | 9999.00 |        |      |
| taxi_distance_raw                           | 0              | 1                  | 580.41         | 422.32 | 0.00       | 0.00    | 888.00 | 888.0   | 888.00 |      |
| bus_trips                                   | 0              | 1                  | 853.67         | 170.30 | 1.00       | 888.00  | 888.00 | 888.0   | 888.00 |      |
| bus_cost_raw                                | 0              | 1                  | 8557.931660.68 | 2.00   | 8888.00    | 8888.00 | 8888.0 | 9999.00 |        |      |
| bus_distance_raw                            | 0              | 1                  | 853.34         | 171.92 | 0.00       | 888.00  | 888.00 | 888.0   | 888.00 |      |
| train_trips                                 | 0              | 1                  | 884.60         | 54.56  | 1.00       | 888.00  | 888.00 | 888.0   | 888.00 |      |
| train_cost_raw                              | 0              | 1                  | 8854.07544.61  | 7.00   | 8888.00    | 8888.00 | 8888.0 | 8888.00 |        |      |
| train_dis-<br>tance_raw                     | 0              | 1                  | 884.57         | 54.99  | 0.00       | 888.00  | 888.00 | 888.0   | 888.00 |      |
| nearest_health_fa-<br>cility_code           | 0              | 1                  | 2.86           | 2.02   | 1.00       | 2.00    | 2.00   | 2.0     | 99.00  |      |
| trans-<br>port_to_health_fa-<br>cility_code | 55             | 1                  | 2.20           | 1.64   | 1.00       | 1.00    | 2.00   | 2.0     | 9.00   |      |
| time_to_health_fa-<br>cility_raw            | 0              | 1                  | 1.80           | 0.83   | -<br>20.00 | 1.00    | 2.00   | 2.0     | 9.00   |      |
| geo_type_code                               | 0              | 1                  | 1.38           | 0.55   | 1.00       | 1.00    | 1.00   | 2.0     | 3.00   |      |
| household_weight                            | 0              | 1                  | 933.68         | 426.29 | 191.85     | 625.38  | 874.59 | 1117.75 | 49.94  |      |

```
head(D11,5)
```

```
# A tibble: 5 x 33
  survey_year province household_id person_number survey_date_raw quarter
  <dbl> <chr> <chr> <chr> <dbl> <dbl>
1 2024 Western Cape 16010072000001~ 01 1032024 1
2 2024 Western Cape 16010088000000~ 01 1022024 1
3 2024 Western Cape 16010088000000~ 02 1022024 1
4 2024 Western Cape 16010088000000~ 01 1022024 1
5 2024 Western Cape 16010088000000~ 02 1022024 1
# i 27 more variables: province_code <dbl>, geo_type_code <dbl>,
# gender_code <dbl>, age <dbl>, population_group_code <dbl>,
# marital_status_code <dbl>, education_level_code <dbl>,
# field_of_study_code <dbl>, education_attendance_code <dbl>,
# employment_status_code <dbl>, expanded_employment_status_code <dbl>,
# unemployment_status_code <dbl>, education_status_code <dbl>,
# neet_code <dbl>, graduate_status_code <dbl>, age_group_code <dbl>, ...
```

skim(D11)

Table 11: Data summary

|                        |        |
|------------------------|--------|
| Name                   | D11    |
| Number of rows         | 170432 |
| Number of columns      | 33     |
| Column type frequency: |        |
| character              | 3      |
| numeric                | 30     |
| Group variables        | None   |

**Variable type: character**

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---------------|-----------|---------------|-----|-----|-------|----------|------------|
| province      | 0         | 1             | 7   | 13  | 0     | 26       | 0          |
| household_id  | 0         | 1             | 18  | 18  | 0     | 36053    | 0          |
| person_number | 0         | 1             | 2   | 2   | 0     | 23       | 0          |

**Variable type: numeric**

| skim_variable         | n_missing | complete_rate | mean       | sd      | p0      | p25     | p50     | p75     | p100    | hist |
|-----------------------|-----------|---------------|------------|---------|---------|---------|---------|---------|---------|------|
| survey_year           | 0         | 1             | 2024.00    | 0.00    | 2024.0  | 2024.00 | 2024.00 | 2024.00 | 2024.00 |      |
| survey_date_raw       | 0         | 1             | 2563464.05 | 2114.96 | 2024.10 | 2024.20 | 2024.30 | 2024.40 | 2024.00 |      |
| quarter               | 0         | 1             | 2.50       | 1.12    | 1.0     | 1.00    | 2.00    | 3.00    | 4.00    |      |
| province_code         | 0         | 1             | 5.27       | 2.55    | 1.0     | 3.00    | 5.00    | 7.00    | 9.00    |      |
| geo_type_code         | 0         | 1             | 1.38       | 0.56    | 1.0     | 1.00    | 1.00    | 2.00    | 3.00    |      |
| gender_code           | 0         | 1             | 1.52       | 0.50    | 1.0     | 1.00    | 2.00    | 2.00    | 2.00    |      |
| age                   | 0         | 1             | 35.72      | 14.01   | -7.0    | 24.00   | 34.00   | 47.00   | 175.00  |      |
| population_group_code | 0         | 1             | 1.30       | 0.75    | 1.0     | 1.00    | 1.00    | 1.00    | 4.00    |      |
| marital_status_code   | 0         | 1             | 3.78       | 1.71    | 1.0     | 2.00    | 5.00    | 5.00    | 5.00    |      |

| skim_variable                             | n_miss-<br>ing | com-<br>plete_rate | mean   | sd     | p0   | p25     | p50    | p75     | p100    | hist |
|-------------------------------------------|----------------|--------------------|--------|--------|------|---------|--------|---------|---------|------|
| educa-<br>tion_level_code                 | 160            | 1                  | 13.85  | 13.10  | 0.0  | 10.00   | 11.00  | 12.00   | 98.00   |      |
| field_of_study_code                       | 0              | 1                  | Inf    | NaN    | 1.0  | Inf     | Inf    | Inf     | Inf     |      |
| education_at-<br>tendance_code            | 0              | 1                  | 1.83   | 0.38   | 1.0  | 2.00    | 2.00   | 2.00    | 2.00    |      |
| employ-<br>ment_sta-<br>tus_code          | 0              | 1                  | 2.39   | 1.31   | 1.0  | 1.00    | 2.00   | 4.00    | 4.00    |      |
| expanded_em-<br>ployment_sta-<br>tus_code | 0              | 1                  | 2.27   | 1.27   | 1.0  | 1.00    | 2.00   | 4.00    | 4.00    |      |
| unemploy-<br>ment_sta-<br>tus_code        | 0              | 1                  | Inf    | NaN    | 1.0  | 3.00    | 5.00   | Inf     | Inf     |      |
| education_sta-<br>tus_code                | 0              | 1                  | 4.36   | 1.11   | 1.0  | 4.00    | 4.00   | 5.00    | 7.00    |      |
| neet_code                                 | 0              | 1                  | 1.55   | 0.50   | 1.0  | 1.00    | 2.00   | 2.00    | 2.00    |      |
| graduate_sta-<br>tus_code                 | 0              | 1                  | 1.65   | 0.90   | 1.0  | 1.00    | 1.00   | 2.00    | 5.00    |      |
| age_group_code                            | 0              | 1                  | 7.75   | 2.78   | 4.0  | 5.00    | 7.00   | 10.00   | 13.00   |      |
| occupa-<br>tion_code                      | 0              | 1                  | Inf    | NaN    | 1.0  | 8.00    | Inf    | Inf     | Inf     |      |
| industry_code                             | 0              | 1                  | Inf    | NaN    | 1.0  | 8.00    | Inf    | Inf     | Inf     |      |
| sector1code                               | 0              | 1                  | Inf    | NaN    | 1.0  | 1.00    | Inf    | Inf     | Inf     |      |
| sector2code                               | 0              | 1                  | Inf    | NaN    | 1.0  | 1.00    | Inf    | Inf     | Inf     |      |
| informal_em-<br>ployment_code             | 0              | 1                  | Inf    | NaN    | 1.0  | 2.00    | Inf    | Inf     | Inf     |      |
| long_term_un-<br>employment_code          | 0              | 1                  | Inf    | NaN    | 1.0  | Inf     | Inf    | Inf     | Inf     |      |
| underemploy-<br>ment_code                 | 0              | 1                  | Inf    | NaN    | 1.0  | 2.00    | Inf    | Inf     | Inf     |      |
| hours_worked_raw                          | 45             | 1                  | Inf    | NaN    | 1.0  | 45.00   | Inf    | Inf     | Inf     |      |
| monthly_amount_raw                        | 0              | 1                  | Inf    | NaN    | 2.0  | 8667.00 | Inf    | Inf     | Inf     |      |
| quarter_weight                            | 0              | 1                  | 970.94 | 468.99 | 50.0 | 667.81  | 906.89 | 1195.88 | 6169.32 |      |
| annual_weight                             | 0              | 1                  | 242.73 | 117.25 | 12.5 | 166.95  | 226.72 | 298.97  | 1542.33 |      |

## Check For missing Data:

Total missing values for each dataset

```
paste0("D01: ", sum(is.na(D01))) #paste0 combines the text with 0 spaces in between
```

```
[1] "D01: 115"
```

```
paste0("D02: ", sum(is.na(D02)))
```

```
[1] "D02: 120"
```

```
paste0("D03: ", sum(is.na(D03)))
```

```
[1] "D03: 100"
```

```
paste0("D04: ", sum(is.na(D04)))
```

```
[1] "D04: 55"
```

```
paste0("D05: ", sum(is.na(D05)))
```

```
[1] "D05: 60"
```

```
paste0("D06: ", sum(is.na(D06)))
```

```
[1] "D06: 50"
```

```
paste0("D07: ", sum(is.na(D07)))
```

```
[1] "D07: 65"
```

```
paste0("D08: ", sum(is.na(D08)))
```

```
[1] "D08: 60"
```

```
paste0("D09: ", sum(is.na(D09)))
```

```
[1] "D09: 65"
```

```
paste0("D10: ", sum(is.na(D10)))
```

```
[1] "D10: 55"
```

```
paste0("D11: ", sum(is.na(D11)))
```

```
[1] "D11: 205"
```

Dataset 11 has the most missing values with 205, with the person level datasets D01-D03 having around 100 and the rest of the datasets that are household level (D04-D10) all have around the 50-60 range.

### Diving deeper into which columns have the missing values in each dataset

```
data.frame(missing = colSums(is.na(D01)))
```

|                           | missing |
|---------------------------|---------|
| survey_year               | 0       |
| province                  | 0       |
| household_id              | 25      |
| person_number             | 0       |
| psu                       | 0       |
| province_code             | 0       |
| sex_code                  | 0       |
| age                       | 90      |
| age_group_code            | 0       |
| population_group_code     | 0       |
| language_code             | 0       |
| relationship_to_head_code | 0       |
| marital_status_code       | 0       |
| geo_type_code             | 0       |
| metro_code                | 0       |
| metro_indicator           | 0       |
| person_weight             | 0       |

There are 2 columns in Dataset 1 with missing values, (household\_id = 25 missing values and age = 90 missing values)

```
data.frame(missing=colSums(is.na(D02)))
```

|                                  | missing |
|----------------------------------|---------|
| survey_year                      | 0       |
| province                         | 0       |
| household_id                     | 0       |
| person_number                    | 0       |
| province_code                    | 0       |
| age                              | 0       |
| sex_code                         | 0       |
| education_level_code             | 120     |
| education_attendance_code        | 0       |
| reason_not_attending_code        | 0       |
| institution_type_code            | 0       |
| public_private_institution_code  | 0       |
| transport_mode_to_education_code | 0       |
| travel_time_to_education         | 0       |
| bursary_code                     | 0       |
| total_education_fees             | 0       |
| no_fees_school_code              | 0       |
| grade_code                       | 0       |
| school_food_programme_code       | 0       |
| absence_code                     | 0       |
| absence_days                     | 0       |
| geo_type_code                    | 0       |
| person_weight                    | 0       |

There is 1 column in Dataset 2 with missing values (education\_level\_code = 120 missing values)

```
data.frame(missing=colSums(is.na(D03)))
```

|               | missing |
|---------------|---------|
| survey_year   | 0       |
| province      | 0       |
| household_id  | 0       |
| person_number | 0       |
| province_code | 0       |

|                        |     |
|------------------------|-----|
| age                    | 0   |
| sex_code               | 0   |
| education_level_code   | 0   |
| wage_work_code         | 0   |
| business_work_code     | 0   |
| volunteer_work_code    | 0   |
| retired_code           | 0   |
| salary_amount_raw      | 0   |
| salary_period_code     | 0   |
| salary_category_code   | 0   |
| transport_to_work_code | 0   |
| minutes_to_work        | 0   |
| sector_code            | 0   |
| looking_for_work_code  | 0   |
| could_start_work_code  | 0   |
| employment_status1code | 100 |
| employment_status2code | 0   |
| lab_salary_raw         | 0   |
| geo_type_code          | 0   |
| person_weight          | 0   |

There is 1 column in Dataset 3 with missing values (employment\_status1code = 100 missing values)

```
data.frame(missing = colSums(is.na(D04)))
```

|                            | missing |
|----------------------------|---------|
| survey_year                | 0       |
| province                   | 0       |
| household_id               | 0       |
| province_code              | 0       |
| head_population_group_code | 0       |
| head_sex_code              | 0       |
| head_age                   | 0       |
| main_dwelling_code         | 0       |
| wall_material_code         | 55      |
| roof_material_code         | 0       |
| floor_material_code        | 0       |
| wall_condition_code        | 0       |
| roof_condition_code        | 0       |
| floor_condition_code       | 0       |
| total_rooms                | 0       |



|                      |   |
|----------------------|---|
| tenure_code          | 0 |
| rent_raw             | 0 |
| rdp_housing_code     | 0 |
| housing_subsidy_code | 0 |
| household_size       | 0 |
| geo_type_code        | 0 |
| metro_code           | 0 |
| household_weight     | 0 |

There is 1 column in Dataset 4 with missing values (wall\_material\_code = 55 missing values)

```
data.frame(missing=colSums(is.na(D05)))
```

|                                   | missing |
|-----------------------------------|---------|
| survey_year                       | 0       |
| province                          | 0       |
| household_id                      | 0       |
| province_code                     | 0       |
| drinking_water_source_code        | 60      |
| distance_to_water_code            | 0       |
| time_to_water_raw                 | 0       |
| pipied_water_code                 | 0       |
| water_safe_code                   | 0       |
| water_clear_code                  | 0       |
| water_taste_code                  | 0       |
| water_smell_code                  | 0       |
| water_treatment_code              | 0       |
| municipal_water_source_code       | 0       |
| water_meter_code                  | 0       |
| pays_for_water_code               | 0       |
| water_interruptions12months_code  | 0       |
| water_interruption_frequency_code | 0       |
| water_alternative_source_code     | 0       |
| main_toilet_code                  | 0       |
| shared_toilet_code                | 0       |
| toilet_location_code              | 0       |
| toilet_distance_code              | 0       |
| handwashing_facility_code         | 0       |
| handwashing_code                  | 0       |
| geo_type_code                     | 0       |
| household_weight                  | 0       |

There is 1 column in Dataset 5 with missing values (drinking\_water\_source\_code = 60 missing values)

```
data.frame(missing=colSums(is.na(D06)))
```

|                              | missing |
|------------------------------|---------|
| survey_year                  | 0       |
| province                     | 0       |
| household_id                 | 0       |
| province_code                | 0       |
| electricity_access_code      | 50      |
| main_electricity_source_code | 0       |
| mains_electricity_code       | 0       |
| electricity_supply_code      | 0       |
| cooking_energy_code          | 0       |
| lighting_energy_code         | 0       |
| space_heating_energy_code    | 0       |
| loadshedding_frequency_code  | 0       |
| alternative_lighting_code    | 0       |
| alternative_cooking_code     | 0       |
| pays_for_electricity_code    | 0       |
| solar_geyser_asset_code      | 0       |
| solar_panel_asset_code       | 0       |
| geo_type_code                | 0       |
| household_weight             | 0       |

There is 1 column in Dataset 6 with missing values (electricity\_access\_code = 50 missing values)

```
data.frame(missing=colSums(is.na(D07)))
```

|                                 | missing |
|---------------------------------|---------|
| survey_year                     | 0       |
| province                        | 0       |
| household_id                    | 0       |
| province_code                   | 0       |
| adult_hunger_code               | 65      |
| child_hunger_code               | 0       |
| worried_food_would_run_out_code | 0       |
| unable_to_eat_healthy_food_code | 0       |
| ate_few_foods_code              | 0       |

|                                    |   |
|------------------------------------|---|
| few_foods_days                     | 0 |
| skipped_meal_code                  | 0 |
| skipped_meal_days                  | 0 |
| ate_less_code                      | 0 |
| ate_less_days                      | 0 |
| ran_out_of_food_code               | 0 |
| ran_out_days                       | 0 |
| hungry_but_did_not_eat_code        | 0 |
| whole_day_without_eating_code      | 0 |
| household_size                     | 0 |
| total_monthly_household_income_raw | 0 |
| geo_type_code                      | 0 |
| household_weight                   | 0 |

There is 1 column in Dataset 7 with missing values (adult\_hunger\_code = 65 missing values)

```
data.frame(missing=colSums(is.na(D08)))
```

|                            | missing |
|----------------------------|---------|
| survey_year                | 0       |
| province                   | 0       |
| household_id               | 0       |
| province_code              | 0       |
| telephone_code             | 0       |
| cellphone_code             | 0       |
| fixed_internet_code        | 0       |
| mobile_internet_code       | 60      |
| public_wi-fi_internet_code | 0       |
| study_internet_code        | 0       |
| work_internet_code         | 0       |
| library_internet_code      | 0       |
| cafe_internet_code         | 0       |
| household_mobile_phones    | 0       |
| household_smartphones      | 0       |
| computer_asset_code        | 0       |
| geo_type_code              | 0       |
| household_weight           | 0       |

There is 1 column in Dataset 8 with missing values (mobile\_internet\_code = 60 missing values)

```
data.frame(missing=colSums(is.na(D09)))
```

|                                    | missing |
|------------------------------------|---------|
| survey_year                        | 0       |
| province                           | 0       |
| household_id                       | 0       |
| province_code                      | 0       |
| income_salary_code                 | 0       |
| income_business_code               | 0       |
| income_remittance_code             | 0       |
| income_pension_code                | 0       |
| income_grant_code                  | 0       |
| income_agriculture_code            | 0       |
| no_income_code                     | 0       |
| other_income_code                  | 0       |
| main_income_source_code            | 65      |
| required_income_raw                | 0       |
| comparative_income_code            | 0       |
| expenditure_band_code              | 0       |
| total_monthly_household_income_raw | 0       |
| vehicle_asset_code                 | 0       |
| total_vehicles                     | 0       |
| tv_asset_code                      | 0       |
| pay_tv_asset_code                  | 0       |
| computer_asset_code                | 0       |
| washing_machine_asset_code         | 0       |
| freezer_asset_code                 | 0       |
| fridge_asset_code                  | 0       |
| microwave_asset_code               | 0       |
| geyser_asset_code                  | 0       |
| solar_panel_asset_code             | 0       |
| radio_asset_code                   | 0       |
| household_size                     | 0       |
| salary_earners_in_household        | 0       |
| social_grant_recipients            | 0       |
| total_grant_recipients             | 0       |
| household_weight                   | 0       |

There is 1 column in Dataset 9 with missing values (main\_income\_source\_code = 65 missing values)

```
data.frame(missing=colSums(is.na(D10)))
```

|                                   | missing |
|-----------------------------------|---------|
| survey_year                       | 0       |
| province                          | 0       |
| household_id                      | 0       |
| province_code                     | 0       |
| taxi_use_code                     | 0       |
| bus_use_code                      | 0       |
| train_use_code                    | 0       |
| taxi_trips                        | 0       |
| taxi_cost_raw                     | 0       |
| taxi_distance_raw                 | 0       |
| bus_trips                         | 0       |
| bus_cost_raw                      | 0       |
| bus_distance_raw                  | 0       |
| train_trips                       | 0       |
| train_cost_raw                    | 0       |
| train_distance_raw                | 0       |
| nearest_health_facility_code      | 0       |
| transport_to_health_facility_code | 55      |
| time_to_health_facility_raw       | 0       |
| geo_type_code                     | 0       |
| household_weight                  | 0       |

There is 1 column in Dataset 10 with missing values (transport\_to\_health\_facility\_code = 55 missing values)

```
data.frame(missing=colSums(is.na(D11)))
```

|                 | missing |
|-----------------|---------|
| survey_year     | 0       |
| province        | 0       |
| household_id    | 0       |
| person_number   | 0       |
| survey_date_raw | 0       |
| quarter         | 0       |
| province_code   | 0       |
| geo_type_code   | 0       |
| gender_code     | 0       |
| age             | 0       |

|                                 |     |
|---------------------------------|-----|
| population_group_code           | 0   |
| marital_status_code             | 0   |
| education_level_code            | 160 |
| field_of_study_code             | 0   |
| education_attendance_code       | 0   |
| employment_status_code          | 0   |
| expanded_employment_status_code | 0   |
| unemployment_status_code        | 0   |
| education_status_code           | 0   |
| neet_code                       | 0   |
| graduate_status_code            | 0   |
| age_group_code                  | 0   |
| occupation_code                 | 0   |
| industry_code                   | 0   |
| sector1code                     | 0   |
| sector2code                     | 0   |
| informal_employment_code        | 0   |
| long_term_unemployment_code     | 0   |
| underemployment_code            | 0   |
| hours_worked_raw                | 45  |
| monthly_amount_raw              | 0   |
| quarter_weight                  | 0   |
| annual_weight                   | 0   |

There are 2 columns in Dataset 11 with missing values (education\_level\_code = 160 missing values and hours\_worked\_raw = 45 missing values)

## Checking for duplicate Rows:

### Total duplicated values for each dataset

```
paste0("D01: ", D01 |>
  get_dupes(household_id, person_number) |>
  nrow())
```

```
[1] "D01: 23"
```

```
paste0("D02: ", D02 |>
  get_dupes(household_id, person_number) |>
  nrow())
```

[1] "D02: 44"

```
paste0("D03: ",D03 |>  
  get_dupes(household_id,person_number) |>  
  nrow())
```

[1] "D03: 40"

```
paste0("D04: ",D04 |>  
  get_dupes(household_id) |>  
  nrow())
```

[1] "D04: 32"

```
paste0("D05: ",D05 |>  
  get_dupes(household_id) |>  
  nrow())
```

[1] "D05: 0"

```
paste0("D06: ",D06 |>  
  get_dupes(household_id) |>  
  nrow())
```

[1] "D06: 0"

```
paste0("D07: ",D07 |>  
  get_dupes(household_id) |>  
  nrow())
```

[1] "D07: 30"

```
paste0("D08: ",D08 |>  
  get_dupes(household_id) |>  
  nrow())
```

[1] "D08: 0"

```
paste0("D09: ",D09 |>
  get_dupes(household_id) |>
  nrow())
```

```
[1] "D09: 30"
```

```
paste0("D10: ",D10 |>
  get_dupes(household_id) |>
  nrow())
```

```
[1] "D10: 0"
```

```
paste0("D11: ",D11 |>
  get_dupes() |>
  nrow())
```

```
[1] "D11: 60"
```

When we look at the duplicated values we see they are all below 100 which is good with most of them even below 50, the only above 50 is Dataset 11 and it has over 170000 rows which means that the data is relatively clean on duplicates.

## Diving deeper into the duplicated values in each dataset

```
D01 |>
  get_dupes(household_id, person_number) #This is a composite key we use on D01,D02,D03 sin
```

```
# A tibble: 23 x 18
```

|   | household_id | person_number | dupe_count | survey_year | province      | psu         |
|---|--------------|---------------|------------|-------------|---------------|-------------|
|   | <chr>        | <chr>         | <int>      | <dbl>       | <chr>         | <dbl>       |
| 1 | <NA>         | 02            | 7          | 2024        | Western Cape  | 17810007000 |
| 2 | <NA>         | 02            | 7          | 2024        | Western Cape  | 19910634000 |
| 3 | <NA>         | 02            | 7          | 2024        | Eastern Cape  | 29410013000 |
| 4 | <NA>         | 02            | 7          | 2024        | Northern Cape | 38310278000 |
| 5 | <NA>         | 02            | 7          | 2024        | Gauteng       | 79811496000 |
| 6 | <NA>         | 02            | 7          | 2024        | Gauteng       | 79912166000 |
| 7 | <NA>         | 02            | 7          | 2024        | Limpopo       | 98710223000 |
| 8 | <NA>         | 01            | 5          | 2024        | Eastern Cape  | 26010501009 |



```

 9 <NA>          01          5      2024 Eastern Cape  29110242000
10 <NA>          01          5      2024 Eastern Cape  29810316000
# i 13 more rows
# i 12 more variables: province_code <dbl>, sex_code <dbl>, age <dbl>,
#   age_group_code <dbl>, population_group_code <dbl>, language_code <dbl>,
#   relationship_to_head_code <dbl>, marital_status_code <dbl>,
#   geo_type_code <dbl>, metro_code <dbl>, metro_indicator <dbl>,
#   person_weight <dbl>

```

Data relatively clean with duplicates for Dataset 1 only duplicates occur due to NA for household\_id

```

D02 |>
  get_dupes(household_id, person_number)

```

```

# A tibble: 44 x 24
  household_id      person_number dupe_count survey_year province province_code
  <chr>           <chr>           <int>    <dbl> <chr>           <dbl>
1 1651013900000297~ 01                2      2024 Western~         1
2 1651013900000297~ 01                2      2024 Western~         1
3 1721011200000230~ 02                2      2024 Western~         1
4 1721011200000230~ 02                2      2024 Western~         1
5 2651001800000150~ 02                2      2024 Eastern~         2
6 2651001800000150~ 02                2      2024 Eastern~         2
7 2941031700000017~ 06                2      2024 Eastern~         2
8 2941031700000017~ 06                2      2024 Eastern~         2
9 2941106100000277~ 03                2      2024 Eastern~         2
10 2941106100000277~ 03                2      2024 Eastern~         2
# i 34 more rows
# i 18 more variables: age <dbl>, sex_code <dbl>, education_level_code <dbl>,
#   education_attendance_code <chr>, reason_not_attending_code <dbl>,
#   institution_type_code <dbl>, public_private_institution_code <dbl>,
#   transport_mode_to_education_code <dbl>, travel_time_to_education <dbl>,
#   bursary_code <dbl>, total_education_fees <dbl>, no_fees_school_code <dbl>,
#   grade_code <dbl>, school_food_programme_code <dbl>, absence_code <dbl>, ...

```

Dataset 2 has 44 rows of duplicate values

```

D03 |>
  get_dupes(household_id, person_number)

```

```
# A tibble: 40 x 26
  household_id person_number dupe_count survey_year province province_code
  <chr>         <chr>          <int>    <dbl> <chr>          <dbl>
1 1991532100000113~ 01              2      2024 Western~      1
2 1991532100000113~ 01              2      2024 Western~      1
3 2601007600000013~ 03              2      2024 Eastern~      2
4 2601007600000013~ 03              2      2024 Eastern~      2
5 2711063900000124~ 02              2      2024 Eastern~      2
6 2711063900000124~ 02              2      2024 Eastern~      2
7 2921009500000144~ 05              2      2024 Eastern~      2
8 2921009500000144~ 05              2      2024 Eastern~      2
9 3611008800000056~ 03              2      2024 Norther~      3
10 3611008800000056~ 03              2      2024 Norther~      3
# i 30 more rows
# i 20 more variables: age <dbl>, sex_code <dbl>, education_level_code <dbl>,
#   wage_work_code <dbl>, business_work_code <dbl>, volunteer_work_code <dbl>,
#   retired_code <dbl>, salary_amount_raw <dbl>, salary_period_code <dbl>,
#   salary_category_code <dbl>, transport_to_work_code <dbl>,
#   minutes_to_work <chr>, sector_code <dbl>, looking_for_work_code <dbl>,
#   could_start_work_code <dbl>, employment_status1code <dbl>, ...
```

Dataset 3 has 40 rows of duplicate values

```
D04 |>
  get_dupes(household_id)
```

```
# A tibble: 32 x 24
  household_id dupe_count survey_year province province_code
  <chr>         <int>    <dbl> <chr>          <dbl>
1 260107750000006601      2      2024 Eastern Cape      2
2 260107750000006601      2      2024 Eastern Cape      2
3 260114010000002901      2      2024 Eastern Cape      2
4 260114010000002901      2      2024 Eastern Cape      2
5 287101060000000501      2      2024 Eastern Cape      2
6 287101060000000501      2      2024 Eastern Cape      2
7 299104250000004101      2      2024 Eastern Cape      2
8 299104250000004101      2      2024 Eastern Cape      2
9 468101440000007901      2      2024 Free State      4
10 468101440000007901      2      2024 Free State      4
# i 22 more rows
# i 19 more variables: head_population_group_code <dbl>, head_sex_code <dbl>,
#   head_age <dbl>, main_dwelling_code <dbl>, wall_material_code <dbl>,
```

```
# roof_material_code <dbl>, floor_material_code <dbl>,
# wall_condition_code <dbl>, roof_condition_code <dbl>,
# floor_condition_code <dbl>, total_rooms <dbl>, tenure_code <dbl>,
# rent_raw <dbl>, rdp_housing_code <dbl>, housing_subsidy_code <dbl>, ...
```

Dataset 4 has 32 rows of duplicate values

```
D05 |>
  get_dupes(household_id)
```

```
# A tibble: 0 x 28
# i 28 variables: household_id <chr>, dupe_count <int>, survey_year <dbl>,
# province <chr>, province_code <dbl>, drinking_water_source_code <dbl>,
# distance_to_water_code <dbl>, time_to_water_raw <chr>,
# piped_water_code <dbl>, water_safe_code <dbl>, water_clear_code <dbl>,
# water_taste_code <dbl>, water_smell_code <dbl>, water_treatment_code <dbl>,
# municipal_water_source_code <dbl>, water_meter_code <dbl>,
# pays_for_water_code <dbl>, water_interruptions12months_code <dbl>, ...
```

Dataset 5 has no duplicate values

```
D06 |>
  get_dupes(household_id)
```

```
# A tibble: 0 x 20
# i 20 variables: household_id <chr>, dupe_count <int>, survey_year <dbl>,
# province <chr>, province_code <dbl>, electricity_access_code <dbl>,
# main_electricity_source_code <dbl>, mains_electricity_code <dbl>,
# electricity_supply_code <dbl>, cooking_energy_code <chr>,
# lighting_energy_code <dbl>, space_heating_energy_code <dbl>,
# loadshedding_frequency_code <dbl>, alternative_lighting_code <dbl>,
# alternative_cooking_code <dbl>, pays_for_electricity_code <dbl>, ...
```

Dataset 6 has no duplicate values

```
D07 |>
  get_dupes(household_id)
```

```
# A tibble: 30 x 23
  household_id dupe_count survey_year province province_code adult_hunger_code
  <chr>         <int>      <dbl> <chr>         <dbl>         <dbl>
1 180101050000~      2      2024 Western~      1             1
2 180101050000~      2      2024 Western~      1             1
3 260108310000~      2      2024 Eastern~      2             3
4 260108310000~      2      2024 Eastern~      2             3
5 284104930000~      2      2024 Eastern~      2             3
6 284104930000~      2      2024 Eastern~      2             3
7 473100190000~      2      2024 Free St~      4             1
8 473100190000~      2      2024 Free St~      4             1
9 499110900000~      2      2024 Free St~      4             1
10 499110900000~      2      2024 Free St~      4             1
# i 20 more rows
# i 17 more variables: child_hunger_code <dbl>,
#   worried_food_would_run_out_code <dbl>,
#   unable_to_eat_healthy_food_code <dbl>, ate_few_foods_code <dbl>,
#   few_foods_days <dbl>, skipped_meal_code <dbl>, skipped_meal_days <dbl>,
#   ate_less_code <dbl>, ate_less_days <dbl>, ran_out_of_food_code <dbl>,
#   ran_out_days <dbl>, hungry_but_did_not_eat_code <dbl>, ...
```

Dataset 6 has no duplicate values

```
D08 |>
  get_dupes(household_id)
```

```
# A tibble: 0 x 19
# i 19 variables: household_id <chr>, dupe_count <int>, survey_year <dbl>,
#   province <chr>, province_code <dbl>, telephone_code <dbl>,
#   cellphone_code <dbl>, fixed_internet_code <dbl>,
#   mobile_internet_code <dbl>, public_wi-fi_internet_code <dbl>,
#   study_internet_code <dbl>, work_internet_code <dbl>,
#   library_internet_code <dbl>, cafe_internet_code <dbl>,
#   household_mobile_phones <dbl>, household_smartphones <dbl>, ...
```

No duplicate rows in Dataset 8

```
D09 |>
  get_dupes(household_id)
```

```
# A tibble: 30 x 35
  household_id dupe_count survey_year province province_code income_salary_code
```

```

      <chr>           <int>      <dbl> <chr>           <dbl>           <dbl>
1 27410248000~      2        2024 Eastern~      2            2
2 27410248000~      2        2024 Eastern~      2            2
3 27410248000~      2        2024 Eastern~      2            2
4 27410248000~      2        2024 Eastern~      2            2
5 28110241000~      2        2024 Eastern~      2            1
6 28110241000~      2        2024 Eastern~      2            1
7 29810319000~      2        2024 Eastern~      2            2
8 29810319000~      2        2024 Eastern~      2            2
9 38010039000~      2        2024 Norther~      3            1
10 38010039000~     2        2024 Norther~      3            1
# i 20 more rows
# i 29 more variables: income_business_code <dbl>,
#   income_remittance_code <dbl>, income_pension_code <dbl>,
#   income_grant_code <dbl>, income_agriculture_code <dbl>,
#   no_income_code <dbl>, other_income_code <dbl>,
#   main_income_source_code <dbl>, required_income_raw <dbl>,
#   comparative_income_code <dbl>, expenditure_band_code <dbl>, ...

```

Dataset 9 has 30 rows of duplicate data

```

D10 |>
  get_dupes(household_id)

```

```

# A tibble: 0 x 22
# i 22 variables: household_id <chr>, dupe_count <int>, survey_year <dbl>,
#   province <chr>, province_code <dbl>, taxi_use_code <dbl>,
#   bus_use_code <dbl>, train_use_code <dbl>, taxi_trips <dbl>,
#   taxi_cost_raw <dbl>, taxi_distance_raw <dbl>, bus_trips <dbl>,
#   bus_cost_raw <dbl>, bus_distance_raw <dbl>, train_trips <dbl>,
#   train_cost_raw <dbl>, train_distance_raw <dbl>,
#   nearest_health_facility_code <dbl>, ...

```

No duplicate rows in Dataset 10

```

D11 |>
  get_dupes()

```

```

# A tibble: 60 x 34
  survey_year province household_id person_number survey_date_raw quarter
  <dbl> <chr>      <chr>      <chr>          <dbl>    <dbl>
1      2024 Eastern Cape 2901009700000~ 02      1032024      1

```

|    |                   |                   |         |   |
|----|-------------------|-------------------|---------|---|
| 2  | 2024 Eastern Cape | 2901009700000~ 02 | 1032024 | 1 |
| 3  | 2024 Eastern Cape | 2991124400000~ 02 | 4112024 | 4 |
| 4  | 2024 Eastern Cape | 2991124400000~ 02 | 4112024 | 4 |
| 5  | 2024 Free State   | 4721060400000~ 01 | 1022024 | 1 |
| 6  | 2024 Free State   | 4721060400000~ 01 | 1022024 | 1 |
| 7  | 2024 Free State   | 4781019900000~ 05 | 3082024 | 3 |
| 8  | 2024 Free State   | 4781019900000~ 05 | 3082024 | 3 |
| 9  | 2024 Gauteng      | 7601104900000~ 04 | 2042024 | 2 |
| 10 | 2024 Gauteng      | 7601104900000~ 04 | 2042024 | 2 |

```
# i 50 more rows
# i 28 more variables: province_code <dbl>, geo_type_code <dbl>,
#   gender_code <dbl>, age <dbl>, population_group_code <dbl>,
#   marital_status_code <dbl>, education_level_code <dbl>,
#   field_of_study_code <dbl>, education_attendance_code <dbl>,
#   employment_status_code <dbl>, expanded_employment_status_code <dbl>,
#   unemployment_status_code <dbl>, education_status_code <dbl>, ...
```

Dataset 11 has 60 rows of duplicate data

## Which Datasets we choose:

D01 - This dataset data about population information about people, for example (age,sex,province,population group) helps us link the other datasets to each other.

D02 - This dataset provides data about education level, this is required to see the level of education attained when we compare it with connectivity level.

D03 - This dataset covers employment data, which is required to see how connectivity compares with job. Chosen over D11 since the row count is similar with consistent id and person number.

D08 - This dataset shows connectivity, this is required to see which households have internet or not it is the core way to determine if there is a link between education, employment and internet connectivity.

## Which Datasets we dont choose:

D09 - Household resources

D04 - Dwelling conditions

D05 - Basic services

D06 - Household energy

D07 - Household wellbeing

D10 - Mobility access

D11 - Labour market

D09 has an affect on whether a household has internet, but we are looking at how internet has an effect being able to get a job or education so for now data set 9 will be skipped but it will be reviewed later if we need more data.

D04,D05,D06,D07,D10,D11 will not be selected for this scenario since they do not greatly affect whether a household has internet, and they also do not help us realize if internet has an affect on education or acquiring a job.

## Check intersect comparison to see if the datasets are compatible with each other:

Now we need to confirm that the datasets we chose are related to each other and can be linked

```
intersect(D01$household_id, D03$household_id) |> length()
```

```
[1] 20940
```

```
intersect(D01$household_id, D02$household_id) |> length()
```

```
[1] 20940
```

```
intersect(D01$household_id, D08$household_id) |> length()
```

```
[1] 20940
```

```
intersect(D01$household_id, D11$household_id) |> length()
```

```
[1] 0
```

We see that for household level, that D01,D02,D03,D08 are very closely related meaning they have the same household ids so they can be used for analysis and comparison

## Check the household ID matching %

```
20684 / n_distinct(D01$household_id) * 100
```

```
[1] 98.77274
```

This tells us that 98.77% of the household ids have a match. This means that out of the total number of household ids in dataset 1, 98.77% of them match with the household ids in data set 2,3 and 8.

**For person level checks we need to use composite keys so we first merge the household\_id and person\_id into a new id variable and then compare those values across datasets:**

```
D01_id <- paste(D01$household_id, D01$person_number)
D02_id <- paste(D02$household_id, D02$person_number)
D03_id <- paste(D03$household_id, D03$person_number)

intersect(D01_id, D02_id) |> length()
```

```
[1] 70415
```

```
intersect(D01_id, D03_id) |> length()
```

```
[1] 70415
```

We see that the values are very close to the total length of the datasets (70000 and change) meaning alot of the composite keys match so we can use this data for comparisons and analysis

D08 is house hold level so we can not compare it at the person level with the others.

## Analysis of our chosen datasets:

### Check for Empty string columns



```
D01 |>
  summarise(across(where(is.character), ~sum(.x == "", na.rm=TRUE)))
```

```
# A tibble: 1 x 3
  province household_id person_number
  <int>      <int>      <int>
1       0          0          0
```

```
D02 |>
  summarise(across(where(is.character), ~sum(.x == "", na.rm=TRUE)))
```

```
# A tibble: 1 x 4
  province household_id person_number education_attendance_code
  <int>      <int>      <int>          <int>
1       0          0          0              0
```

```
D03 |>
  summarise(across(where(is.character), ~sum(.x == "", na.rm=TRUE)))
```

```
# A tibble: 1 x 4
  province household_id person_number minutes_to_work
  <int>      <int>      <int>          <int>
1       0          0          0              0
```

```
D08 |>
  summarise(across(where(is.character), ~sum(.x == "", na.rm=TRUE)))
```

```
# A tibble: 1 x 2
  province household_id
  <int>      <int>
1       0          0
```

We see that there are no empty strings in any of our selected datasets

## Check for Negative values in age

```
paste0("D01: ", sum(D01$age < 0, na.rm = TRUE)) #na.rm means should NA values be counted or not
```

```
[1] "D01: 9"
```

```
paste0("D02: ", sum(D02$age < 0, na.rm = TRUE))
```

```
[1] "D02: 0"
```

```
paste0("D03: ", sum(D03$age < 0, na.rm = TRUE))
```

```
[1] "D03: 0"
```

```
which(D01$age < 0) #Which is used to see which row numbers are below 0 for age
```

```
[1] 2485 6759 15816 17017 17396 19443 40291 62304 63617
```

```
D01 |>
  filter(age < 0)
```

```
# A tibble: 9 x 17
  survey_year province household_id person_number    psu province_code sex_code
    <dbl> <chr>    <chr>    <chr>    <dbl>    <dbl>    <dbl>
1      2024 Western~ 19910583000~ 05      1.99e10      1      2
2      2024 Eastern~ 26010229000~ 06      2.60e10      2      1
3      2024 Eastern~ 29912145000~ 02      2.99e10      2      2
4      2024 Norther~ 37310011000~ 03      3.73e10      3      2
5      2024 Norther~ 37910043000~ 02      3.79e10      3      1
6      2024 Free St~ 46710449000~ 05      4.67e10      4      2
7      2024 Gauteng 76011169000~ 10      7.60e10      7      1
8      2024 Limpopo 96010152000~ 02      9.60e10      9      1
9      2024 Limpopo 96310259000~ 02      9.63e10      9      1
# i 10 more variables: age <dbl>, age_group_code <dbl>,
#   population_group_code <dbl>, language_code <dbl>,
#   relationship_to_head_code <dbl>, marital_status_code <dbl>,
#   geo_type_code <dbl>, metro_code <dbl>, metro_indicator <dbl>,
#   person_weight <dbl>
```

## Check for invalid values in the columns

### Dataset 1:

```
paste0("province_code: ", sum(!D01$province_code %in% c(1,2,3,4,5,6,7,8,9), na.rm = TRUE))
```

```
[1] "province_code: 0"
```

```
paste0("sex_code: ", sum(!D01$sex_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "sex_code: 0"
```

```
paste0("age_group_code: ", sum(!D01$age_group_code %in% c(01,02,03,04,05,06,07,08,09,10,11,12), na.rm = TRUE))
```

```
[1] "age_group_code: 0"
```

```
paste0("population_group_code: ", sum(!D01$population_group_code %in% c(1,2,3,4,5), na.rm = TRUE))
```

```
[1] "population_group_code: 0"
```

```
paste0("language_code: ", sum(!D01$language_code %in% c(01,02,03,04,05,06,07,08,09,10,11,12,13), na.rm = TRUE))
```

```
[1] "language_code: 0"
```

```
paste0("relationship_to_head_code: ", sum(!D01$relationship_to_head_code %in% c(1,2,3,4,5,6,7,8,9), na.rm = TRUE))
```

```
[1] "relationship_to_head_code: 0"
```

```
paste0("marital_status_code: ", sum(!D01$marital_status_code %in% c(1,2,3,4,5,6,7,8,9), na.rm = TRUE))
```

```
[1] "marital_status_code: 0"
```

```
paste0("geo_type_code: ", sum(!D01$geo_type_code %in% c(1,2,3), na.rm = TRUE))
```

```
[1] "geo_type_code: 0"
```

```
paste0("metro_code: ", sum(!D01$metro_code %in% c(01,02,03,04,05,06,07,08,09,10,11,12,13,14,
```

```
[1] "metro_code: 0"
```

```
paste0("metro_indicator: ", sum(!D01$metro_indicator %in% c(1,2), na.rm = TRUE))
```

```
[1] "metro_indicator: 0"
```

## Dataset 2:

```
paste0("province_code: ", sum(!D02$province_code %in% c(1,2,3,4,5,6,7,8,9), na.rm = TRUE))
```

```
[1] "province_code: 0"
```

```
paste0("sex_code: ", sum(!D02$sex_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "sex_code: 0"
```

```
paste0("education_level_code: ", sum(!D02$education_level_code %in% c(98,00,01,02,03,04,05,0
```

```
[1] "education_level_code: 173"
```

```
paste0("education_attendance_code: ", sum(!D02$education_attendance_code %in% c(1,2,3,8), na
```

```
[1] "education_attendance_code: 35"
```

```
paste0("reason_not_attending_code: ", sum(!D02$reason_not_attending_code %in% c(01,02,03,04,
```

```
[1] "reason_not_attending_code: 0"
```

```
paste0("institution_type_code: ", sum(!D02$institution_type_code %in% c(1,2,3,4,5,6,7,8,9,88,
```

```
[1] "institution_type_code: 0"
```

```
paste0("public_private_institution_code: ", sum(!D02$public_private_institution_code %in% c(
```

```
[1] "public_private_institution_code: 0"
```

```
paste0("transport_mode_to_education_code: ", sum(!D02$transport_mode_to_education_code %in% c(
```

```
[1] "transport_mode_to_education_code: 0"
```

```
paste0("travel_time_to_education: ", sum(!D02$travel_time_to_education %in% c(1,2,3,4,5,6,7,8
```

```
[1] "travel_time_to_education: 0"
```

```
paste0("total_education_fees: ", sum(!D02$total_education_fees %in% c(00,01,02,03,04,05,06,07
```

```
[1] "total_education_fees: 0"
```

```
paste0("no_fees_school_code: ", sum(!D02$no_fees_school_code %in% c(1,2,3,4,5,6,7,8), na.rm =
```

```
[1] "no_fees_school_code: 0"
```

```
paste0("grade_code: ", sum(!D02$grade_code %in% c(00,01,02,03,04,05,06,07,08,09,10,11,12,13,14
```

```
[1] "grade_code: 0"
```

```
paste0("school_food_programme_code: ", sum(!D02$school_food_programme_code %in% c(1,2,3,8), na
```

```
[1] "school_food_programme_code: 0"
```

```
paste0("absence_code: ", sum(!D02$absence_code %in% c(1,2,3,4,8), na.rm = TRUE))
```

```
[1] "absence_code: 0"
```

```
paste0("absence_days: ", sum(!D02$absence_days %in% c(1,2,3,4,5,8), na.rm = TRUE))
```

```
[1] "absence_days: 0"
```

```
paste0("geo_type_code: ", sum(!D02$geo_type_code %in% c(1,2,3), na.rm = TRUE))
```

```
[1] "geo_type_code: 0"
```

## Exploring the invalid values we have in columns

```
D02 |>
```

```
  filter(!D02$education_level_code %in% c(98,00,01,02,03,04,05,06,07,08,09,10,11,12,13,14,15
```

```
# A tibble: 173 x 23
```

|    | survey_year | province | household_id   | person_number | province_code | age   | sex_code |
|----|-------------|----------|----------------|---------------|---------------|-------|----------|
|    | <dbl>       | <chr>    | <chr>          | <chr>         | <dbl>         | <dbl> | <dbl>    |
| 1  | 2024        | Western  | ~ 16310232000~ | 03            | 1             | 18    | 2        |
| 2  | 2024        | Western  | ~ 16610155000~ | 02            | 1             | 67    | 2        |
| 3  | 2024        | Western  | ~ 16810060000~ | 01            | 1             | 44    | 1        |
| 4  | 2024        | Western  | ~ 17210065000~ | 03            | 1             | 10    | 2        |
| 5  | 2024        | Western  | ~ 17510156000~ | 04            | 1             | 2     | 2        |
| 6  | 2024        | Western  | ~ 17710299000~ | 01            | 1             | 59    | 1        |
| 7  | 2024        | Western  | ~ 19910370000~ | 06            | 1             | 11    | 1        |
| 8  | 2024        | Western  | ~ 19911516000~ | 03            | 1             | 7     | 1        |
| 9  | 2024        | Western  | ~ 19911801000~ | 05            | 1             | 21    | 1        |
| 10 | 2024        | Western  | ~ 19912454000~ | 02            | 1             | 21    | 1        |

```
# i 163 more rows
```

```
# i 16 more variables: education_level_code <dbl>,
```

```
# education_attendance_code <chr>, reason_not_attending_code <dbl>,
```

```
# institution_type_code <dbl>, public_private_institution_code <dbl>,
```

```
# transport_mode_to_education_code <dbl>, travel_time_to_education <dbl>,
```

```
# bursary_code <dbl>, total_education_fees <dbl>, no_fees_school_code <dbl>,
```

```
# grade_code <dbl>, school_food_programme_code <dbl>, absence_code <dbl>, ...
```

```
D02 |>
```

```
  filter(!D02$education_attendance_code %in% c(1,2,3,8))
```

```
# A tibble: 35 x 23
```

|   | survey_year | province | household_id   | person_number | province_code | age   | sex_code |
|---|-------------|----------|----------------|---------------|---------------|-------|----------|
|   | <dbl>       | <chr>    | <chr>          | <chr>         | <dbl>         | <dbl> | <dbl>    |
| 1 | 2024        | Western  | ~ 19910819000~ | 01            | 1             | 52    | 1        |
| 2 | 2024        | Western  | ~ 19912556000~ | 02            | 1             | 72    | 1        |
| 3 | 2024        | Western  | ~ 19915811000~ | 02            | 1             | 53    | 1        |

|    |                                |   |    |   |
|----|--------------------------------|---|----|---|
| 4  | 2024 Eastern ~ 26011173000~ 01 | 2 | 46 | 2 |
| 5  | 2024 Eastern ~ 27010335000~ 05 | 2 | 4  | 2 |
| 6  | 2024 Eastern ~ 27110629000~ 01 | 2 | 56 | 2 |
| 7  | 2024 Eastern ~ 27410265000~ 04 | 2 | 0  | 1 |
| 8  | 2024 Eastern ~ 28610111000~ 01 | 2 | 57 | 2 |
| 9  | 2024 Eastern ~ 29510413000~ 01 | 2 | 55 | 1 |
| 10 | 2024 Eastern ~ 29911334000~ 01 | 2 | 47 | 2 |

```
# i 25 more rows
# i 16 more variables: education_level_code <dbl>,
#   education_attendance_code <chr>, reason_not_attending_code <dbl>,
#   institution_type_code <dbl>, public_private_institution_code <dbl>,
#   transport_mode_to_education_code <dbl>, travel_time_to_education <dbl>,
#   bursary_code <dbl>, total_education_fees <dbl>, no_fees_school_code <dbl>,
#   grade_code <dbl>, school_food_programme_code <dbl>, absence_code <dbl>, ...
```

### Dataset 3:

```
paste0("province_code: ", sum(!D03$province_code %in% c(1,2,3,4,5,6,7,8,9), na.rm = TRUE))
```

```
[1] "province_code: 0"
```

```
paste0("sex_code: ", sum(!D03$sex_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "sex_code: 0"
```

```
paste0("education_level_code: ", sum(!D03$education_level_code %in% c(98,00,01,02,03,04,05,06,07,08,09,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,00), na.rm = TRUE))
```

```
[1] "education_level_code: 53"
```

```
paste0("wage_work_code: ", sum(!D03$wage_work_code %in% c(1,2,8), na.rm = TRUE))
```

```
[1] "wage_work_code: 2"
```

```
paste0("business_work_code: ", sum(!D03$business_work_code %in% c(1,2,8), na.rm = TRUE))
```

```
[1] "business_work_code: 2"
```

```
paste0("volunteer_work_code: ", sum(!D03$volunteer_work_code %in% c(1,2,8), na.rm = TRUE))
```

```
[1] "volunteer_work_code: 2"
```

```
paste0("retired_code: ", sum(!D03$retired_code %in% c(1,2,8), na.rm = TRUE))
```

```
[1] "retired_code: 0"
```

```
paste0("salary_amount_raw: ", sum(D03$salary_amount_raw < 0, na.rm = TRUE))
```

```
[1] "salary_amount_raw: 0"
```

```
paste0("salary_period_code: ", sum(!D03$salary_period_code %in% c(1,2,3,8), na.rm = TRUE))
```

```
[1] "salary_period_code: 0"
```

```
paste0("salary_category_code: ", sum(!D03$salary_category_code %in% c(1,2,3,4,5,6,7,8,9,10,11), na.rm = TRUE))
```

```
[1] "salary_category_code: 0"
```

```
paste0("transport_to_work_code: ", sum(!D03$transport_to_work_code %in% c(1,2,3,4,5,6,7,8,9,10,11), na.rm = TRUE))
```

```
[1] "transport_to_work_code: 0"
```

```
paste0("minutes_to_work: ", sum(!D03$minutes_to_work %in% c(1,2,3,4,5,6,8), na.rm = TRUE))
```

```
[1] "minutes_to_work: 30"
```

```
paste0("sector_code: ", sum(!D03$sector_code %in% c(1,2,3,8), na.rm = TRUE))
```

```
[1] "sector_code: 0"
```

```
paste0("looking_for_work_code: ", sum(!D03$looking_for_work_code %in% c(1,2,8), na.rm = TRUE))
```

```
[1] "looking_for_work_code: 0"
```



```
paste0("could_start_work_code: ", sum(!D03$could_start_work_code %in% c(1,2,3,8), na.rm = TRUE))
```

```
[1] "could_start_work_code: 0"
```

```
paste0("employment_status1code: ", sum(!D03$employment_status1code %in% c(1,2,8,9), na.rm = TRUE))
```

```
[1] "employment_status1code: 100"
```

```
paste0("employment_status2code: ", sum(!D03$employment_status2code %in% c(1,2,3,8), na.rm = TRUE))
```

```
[1] "employment_status2code: 0"
```

```
paste0("lab_salary_raw: ", sum(D03$lab_salary_raw < 0, na.rm = TRUE))
```

```
[1] "lab_salary_raw: 0"
```

## Exploring the invalid values we have in columns

```
D03 |>
```

```
  filter(!D03$education_level_code %in% c(98,00,01,02,03,04,05,06,07,08,09,10,11,12,13,14,15))
```

```
# A tibble: 53 x 25
```

|    | survey_year | province | household_id   | person_number | province_code | age   | sex_code |
|----|-------------|----------|----------------|---------------|---------------|-------|----------|
|    | <dbl>       | <chr>    | <chr>          | <chr>         | <dbl>         | <dbl> | <dbl>    |
| 1  | 2024        | Western  | ~ 16310232000~ | 03            | 1             | 18    | 2        |
| 2  | 2024        | Western  | ~ 17210065000~ | 03            | 1             | 10    | 2        |
| 3  | 2024        | Western  | ~ 19910370000~ | 06            | 1             | 11    | 1        |
| 4  | 2024        | Western  | ~ 19911516000~ | 03            | 1             | 7     | 1        |
| 5  | 2024        | Western  | ~ 19911801000~ | 05            | 1             | 21    | 1        |
| 6  | 2024        | Western  | ~ 19912454000~ | 02            | 1             | 21    | 1        |
| 7  | 2024        | Western  | ~ 19912489000~ | 04            | 1             | 16    | 1        |
| 8  | 2024        | Western  | ~ 19912552000~ | 02            | 1             | 16    | 2        |
| 9  | 2024        | Western  | ~ 19913331000~ | 04            | 1             | 9     | 2        |
| 10 | 2024        | Western  | ~ 19914132000~ | 04            | 1             | 8     | 2        |

```
# i 43 more rows
```

```
# i 18 more variables: education_level_code <dbl>, wage_work_code <dbl>,  
# business_work_code <dbl>, volunteer_work_code <dbl>, retired_code <dbl>,  
# salary_amount_raw <dbl>, salary_period_code <dbl>,
```

```
# salary_category_code <dbl>, transport_to_work_code <dbl>,
# minutes_to_work <chr>, sector_code <dbl>, looking_for_work_code <dbl>,
# could_start_work_code <dbl>, employment_status1code <dbl>, ...
```

```
D03 |>
  filter(!D03$wage_work_code %in% c(1,2,8))
```

```
# A tibble: 2 x 25
  survey_year province household_id person_number province_code age sex_code
    <dbl> <chr>      <chr>      <chr>          <dbl> <dbl>   <dbl>
1      2024 Free State 46710577000~ 01             4    44     1
2      2024 Gauteng   79713838000~ 02             7    57     1
# i 18 more variables: education_level_code <dbl>, wage_work_code <dbl>,
# business_work_code <dbl>, volunteer_work_code <dbl>, retired_code <dbl>,
# salary_amount_raw <dbl>, salary_period_code <dbl>,
# salary_category_code <dbl>, transport_to_work_code <dbl>,
# minutes_to_work <chr>, sector_code <dbl>, looking_for_work_code <dbl>,
# could_start_work_code <dbl>, employment_status1code <dbl>,
# employment_status2code <dbl>, lab_salary_raw <dbl>, ...
```

```
D03 |>
  filter(!D03$business_work_code %in% c(1,2,8))
```

```
# A tibble: 2 x 25
  survey_year province household_id person_number province_code age sex_code
    <dbl> <chr>      <chr>      <chr>          <dbl> <dbl>   <dbl>
1      2024 Free State 46710577000~ 01             4    44     1
2      2024 Gauteng   79713838000~ 02             7    57     1
# i 18 more variables: education_level_code <dbl>, wage_work_code <dbl>,
# business_work_code <dbl>, volunteer_work_code <dbl>, retired_code <dbl>,
# salary_amount_raw <dbl>, salary_period_code <dbl>,
# salary_category_code <dbl>, transport_to_work_code <dbl>,
# minutes_to_work <chr>, sector_code <dbl>, looking_for_work_code <dbl>,
# could_start_work_code <dbl>, employment_status1code <dbl>,
# employment_status2code <dbl>, lab_salary_raw <dbl>, ...
```

```
D03 |>
  filter(!D03$volunteer_work_code %in% c(1,2,8))
```

```
# A tibble: 2 x 25
```

```

  survey_year province household_id person_number province_code age sex_code
    <dbl> <chr>      <chr>      <chr>      <dbl> <dbl>      <dbl>
1      2024 Free State 46710577000~ 01          4      44          1
2      2024 Gauteng   79713838000~ 02          7      57          1
# i 18 more variables: education_level_code <dbl>, wage_work_code <dbl>,
#   business_work_code <dbl>, volunteer_work_code <dbl>, retired_code <dbl>,
#   salary_amount_raw <dbl>, salary_period_code <dbl>,
#   salary_category_code <dbl>, transport_to_work_code <dbl>,
#   minutes_to_work <chr>, sector_code <dbl>, looking_for_work_code <dbl>,
#   could_start_work_code <dbl>, employment_status1code <dbl>,
#   employment_status2code <dbl>, lab_salary_raw <dbl>, ...

```

D03 |>

```
filter(!D03$minutes_to_work %in% c(1,2,3,4,5,6,8))
```

# A tibble: 30 x 25

```

  survey_year province household_id person_number province_code age sex_code
    <dbl> <chr>      <chr>      <chr>      <dbl> <dbl>      <dbl>
1      2024 Western ~ 16610124000~ 01          1      46          2
2      2024 Western ~ 16610342000~ 03          1      23          2
3      2024 Western ~ 19916198000~ 02          1      37          2
4      2024 Eastern ~ 26011606000~ 01          2      45          1
5      2024 Eastern ~ 27910085000~ 03          2       5          2
6      2024 Eastern ~ 29810316000~ 06          2      18          1
7      2024 Northern~ 36410029000~ 04          3       6          1
8      2024 KwaZulu~ 56810094000~ 01          5      34          2
9      2024 KwaZulu~ 58210026000~ 03          5      37          2
10     2024 KwaZulu~ 59910023000~ 02          5      38          1
# i 20 more rows
# i 18 more variables: education_level_code <dbl>, wage_work_code <dbl>,
#   business_work_code <dbl>, volunteer_work_code <dbl>, retired_code <dbl>,
#   salary_amount_raw <dbl>, salary_period_code <dbl>,
#   salary_category_code <dbl>, transport_to_work_code <dbl>,
#   minutes_to_work <chr>, sector_code <dbl>, looking_for_work_code <dbl>,
#   could_start_work_code <dbl>, employment_status1code <dbl>, ...

```

D03 |>

```
filter(!D03$employment_status1code %in% c(1,2,8,9))
```

# A tibble: 100 x 25

```

  survey_year province household_id person_number province_code age sex_code

```

|    | <dbl> | <chr> | <chr>   | <chr>             | <dbl> | <dbl> | <dbl> |
|----|-------|-------|---------|-------------------|-------|-------|-------|
| 1  |       | 2024  | Western | ~ 16210039000~ 02 | 1     | 63    | 2     |
| 2  |       | 2024  | Western | ~ 16610031000~ 03 | 1     | 16    | 2     |
| 3  |       | 2024  | Western | ~ 16610121000~ 01 | 1     | 45    | 2     |
| 4  |       | 2024  | Western | ~ 16610294000~ 04 | 1     | 10    | 2     |
| 5  |       | 2024  | Western | ~ 17710202000~ 06 | 1     | 3     | 1     |
| 6  |       | 2024  | Western | ~ 19910166000~ 02 | 1     | 42    | 2     |
| 7  |       | 2024  | Western | ~ 19910774000~ 02 | 1     | 47    | 2     |
| 8  |       | 2024  | Western | ~ 19911179000~ 01 | 1     | 56    | 1     |
| 9  |       | 2024  | Western | ~ 19911478000~ 01 | 1     | 70    | 2     |
| 10 |       | 2024  | Western | ~ 19911753000~ 01 | 1     | 42    | 1     |

# i 90 more rows

# i 18 more variables: education\_level\_code <dbl>, wage\_work\_code <dbl>,  
# business\_work\_code <dbl>, volunteer\_work\_code <dbl>, retired\_code <dbl>,  
# salary\_amount\_raw <dbl>, salary\_period\_code <dbl>,  
# salary\_category\_code <dbl>, transport\_to\_work\_code <dbl>,  
# minutes\_to\_work <chr>, sector\_code <dbl>, looking\_for\_work\_code <dbl>,  
# could\_start\_work\_code <dbl>, employment\_status1code <dbl>, ...

## Dataset 8:

```
paste0("province_code: ", sum(!D08$province_code %in% c(1,2,3,4,5,6,7,8,9), na.rm = TRUE))
```

```
[1] "province_code: 0"
```

```
paste0("telephone_code: ", sum(!D08$telephone_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "telephone_code: 2"
```

```
paste0("cellphone_code: ", sum(!D08$cellphone_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "cellphone_code: 0"
```

```
paste0("fixed_internet_code: ", sum(!D08$fixed_internet_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "fixed_internet_code: 2"
```

```
paste0("mobile_internet_code: ", sum(!D08$mobile_internet_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "mobile_internet_code: 62"
```

```
paste0("public_wi-fi_internet_code: ", sum(!D08$public_wi-fi_internet_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "public_wi-fi_internet_code: 2"
```

```
paste0("study_internet_code: ", sum(!D08$study_internet_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "study_internet_code: 2"
```

```
paste0("work_internet_code: ", sum(!D08$work_internet_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "work_internet_code: 2"
```

```
paste0("library_internet_code: ", sum(!D08$library_internet_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "library_internet_code: 2"
```

```
paste0("cafe_internet_code: ", sum(!D08$cafe_internet_code %in% c(1,2), na.rm = TRUE))
```

```
[1] "cafe_internet_code: 2"
```

```
paste0("household_mobile_phones: ", sum(!D08$household_mobile_phones %in% c(0,1,2,3,4,5,6,7,8,9)))
```

```
[1] "household_mobile_phones: 0"
```

```
paste0("household_smartphones: ", sum(!D08$household_smartphones %in% c(0,1,2,3,4,5,6,7,8,9)))
```

```
[1] "household_smartphones: 15"
```

```
paste0("computer_asset_code: ", sum(!D08$computer_asset_code %in% c(1,2,9), na.rm = TRUE))
```

```
[1] "computer_asset_code: 0"
```

```
paste0("geo_type_code: ", sum(!D08$geo_type_code %in% c(1,2,3), na.rm = TRUE))
```

```
[1] "geo_type_code: 0"
```

### Exploring the invalid values we have in columns

```
D08 |>
  filter(!D08$telephone_code %in% c(1,2))
```

```
# A tibble: 2 x 18
  survey_year province household_id province_code telephone_code cellphone_code
      <dbl> <chr>      <chr>          <dbl>         <dbl>         <dbl>
1      2024 Free Sta~ 47210284000~         4             9             1
2      2024 Gauteng  79713838000~         7             9             1
# i 12 more variables: fixed_internet_code <dbl>, mobile_internet_code <dbl>,
#   public_wi-fi_internet_code <dbl>, study_internet_code <dbl>,
#   work_internet_code <dbl>, library_internet_code <dbl>,
#   cafe_internet_code <dbl>, household_mobile_phones <dbl>,
#   household_smartphones <dbl>, computer_asset_code <dbl>,
#   geo_type_code <dbl>, household_weight <dbl>
```

```
D08 |>
  filter(!D08$fixed_internet_code %in% c(1,2))
```

```
# A tibble: 2 x 18
  survey_year province household_id province_code telephone_code cellphone_code
      <dbl> <chr>      <chr>          <dbl>         <dbl>         <dbl>
1      2024 Free Sta~ 47210284000~         4             9             1
2      2024 Gauteng  79713838000~         7             9             1
# i 12 more variables: fixed_internet_code <dbl>, mobile_internet_code <dbl>,
#   public_wi-fi_internet_code <dbl>, study_internet_code <dbl>,
#   work_internet_code <dbl>, library_internet_code <dbl>,
#   cafe_internet_code <dbl>, household_mobile_phones <dbl>,
#   household_smartphones <dbl>, computer_asset_code <dbl>,
#   geo_type_code <dbl>, household_weight <dbl>
```

```
D08 |>
  filter(!D08$mobile_internet_code %in% c(1,2))
```

```
# A tibble: 62 x 18
  survey_year province household_id province_code telephone_code cellphone_code
      <dbl> <chr>      <chr>          <dbl>          <dbl>          <dbl>
1      2024 Western~ 17210002000~          1            1            1
2      2024 Western~ 19910607000~          1            2            1
3      2024 Western~ 19912489000~          1            2            1
4      2024 Western~ 19912560000~          1            1            1
5      2024 Western~ 19913050000~          1            2            1
6      2024 Western~ 19914019000~          1            1            1
7      2024 Eastern~ 26010831000~          2            2            1
8      2024 Eastern~ 26011191000~          2            2            1
9      2024 Eastern~ 26510076000~          2            2            1
10     2024 Eastern~ 27310333000~          2            2            1
# i 52 more rows
# i 12 more variables: fixed_internet_code <dbl>, mobile_internet_code <dbl>,
#   public_wi-fi_internet_code <dbl>, study_internet_code <dbl>,
#   work_internet_code <dbl>, library_internet_code <dbl>,
#   cafe_internet_code <dbl>, household_mobile_phones <dbl>,
#   household_smartphones <dbl>, computer_asset_code <dbl>,
#   geo_type_code <dbl>, household_weight <dbl>
```

```
D08 |>
  filter(!D08$public_wi-fi_internet_code %in% c(1,2))
```

```
# A tibble: 2 x 18
  survey_year province household_id province_code telephone_code cellphone_code
      <dbl> <chr>      <chr>          <dbl>          <dbl>          <dbl>
1      2024 Free Sta~ 47210284000~          4            9            1
2      2024 Gauteng   79713838000~          7            9            1
# i 12 more variables: fixed_internet_code <dbl>, mobile_internet_code <dbl>,
#   public_wi-fi_internet_code <dbl>, study_internet_code <dbl>,
#   work_internet_code <dbl>, library_internet_code <dbl>,
#   cafe_internet_code <dbl>, household_mobile_phones <dbl>,
#   household_smartphones <dbl>, computer_asset_code <dbl>,
#   geo_type_code <dbl>, household_weight <dbl>
```

```
D08 |>
  filter(!D08$study_internet_code %in% c(1,2))
```

```
# A tibble: 2 x 18
  survey_year province household_id province_code telephone_code cellphone_code
```

```

      <dbl> <chr>      <chr>                <dbl>          <dbl>          <dbl>
1      2024 Free Sta~ 47210284000~            4              9              1
2      2024 Gauteng   79713838000~            7              9              1
# i 12 more variables: fixed_internet_code <dbl>, mobile_internet_code <dbl>,
#   public_wi-fi_internet_code <dbl>, study_internet_code <dbl>,
#   work_internet_code <dbl>, library_internet_code <dbl>,
#   cafe_internet_code <dbl>, household_mobile_phones <dbl>,
#   household_smartphones <dbl>, computer_asset_code <dbl>,
#   geo_type_code <dbl>, household_weight <dbl>

```

```

D08 |>
  filter(!D08$work_internet_code %in% c(1,2))

```

```

# A tibble: 2 x 18
  survey_year province household_id province_code telephone_code cellphone_code
      <dbl> <chr>      <chr>                <dbl>          <dbl>          <dbl>
1      2024 Free Sta~ 47210284000~            4              9              1
2      2024 Gauteng   79713838000~            7              9              1
# i 12 more variables: fixed_internet_code <dbl>, mobile_internet_code <dbl>,
#   public_wi-fi_internet_code <dbl>, study_internet_code <dbl>,
#   work_internet_code <dbl>, library_internet_code <dbl>,
#   cafe_internet_code <dbl>, household_mobile_phones <dbl>,
#   household_smartphones <dbl>, computer_asset_code <dbl>,
#   geo_type_code <dbl>, household_weight <dbl>

```

```

D08 |>
  filter(!D08$library_internet_code %in% c(1,2))

```

```

# A tibble: 2 x 18
  survey_year province household_id province_code telephone_code cellphone_code
      <dbl> <chr>      <chr>                <dbl>          <dbl>          <dbl>
1      2024 Free Sta~ 47210284000~            4              9              1
2      2024 Gauteng   79713838000~            7              9              1
# i 12 more variables: fixed_internet_code <dbl>, mobile_internet_code <dbl>,
#   public_wi-fi_internet_code <dbl>, study_internet_code <dbl>,
#   work_internet_code <dbl>, library_internet_code <dbl>,
#   cafe_internet_code <dbl>, household_mobile_phones <dbl>,
#   household_smartphones <dbl>, computer_asset_code <dbl>,
#   geo_type_code <dbl>, household_weight <dbl>

```



```
D08 |>
  filter(!D08$cafe_internet_code %in% c(1,2))
```

```
# A tibble: 2 x 18
  survey_year province household_id province_code telephone_code cellphone_code
      <dbl> <chr>      <chr>          <dbl>          <dbl>          <dbl>
1      2024 Free Sta~ 47210284000~          4            9            1
2      2024 Gauteng  79713838000~          7            9            1
# i 12 more variables: fixed_internet_code <dbl>, mobile_internet_code <dbl>,
#   public_wi-fi_internet_code <dbl>, study_internet_code <dbl>,
#   work_internet_code <dbl>, library_internet_code <dbl>,
#   cafe_internet_code <dbl>, household_mobile_phones <dbl>,
#   household_smartphones <dbl>, computer_asset_code <dbl>,
#   geo_type_code <dbl>, household_weight <dbl>
```

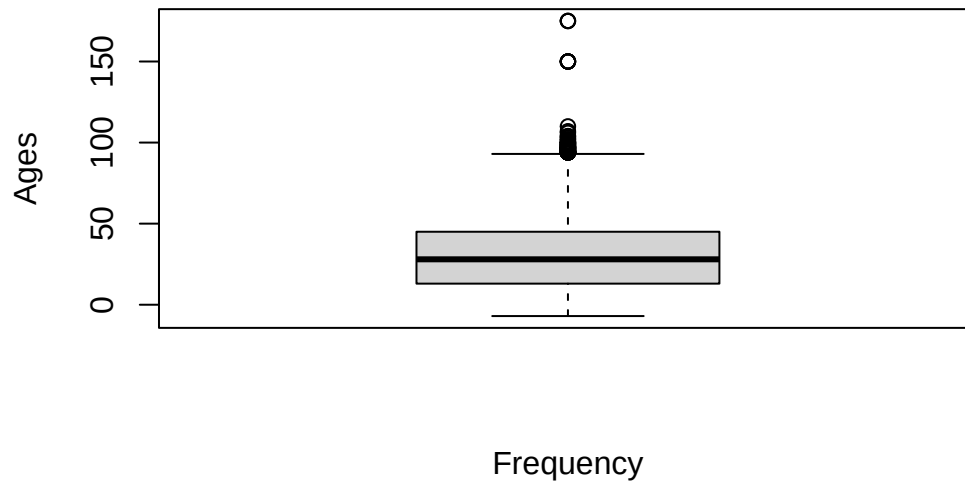
```
D08 |>
  filter(!D08$household_smartphones %in% c(0,1,2,3,4,5,6,7,8,9,10,11,12,13))
```

```
# A tibble: 15 x 18
  survey_year province household_id province_code telephone_code cellphone_code
      <dbl> <chr>      <chr>          <dbl>          <dbl>          <dbl>
1      2024 Norther~ 38610042000~          3            2            1
2      2024 Free St~ 46410077000~          4            2            1
3      2024 KwaZulu~ 52410312000~          5            2            1
4      2024 KwaZulu~ 56710084000~          5            2            1
5      2024 KwaZulu~ 58210252000~          5            2            1
6      2024 KwaZulu~ 59920642000~          5            2            1
7      2024 Gauteng  76210130000~          7            2            1
8      2024 Gauteng  79710804000~          7            2            1
9      2024 Gauteng  79713673051~          7            2            1
10     2024 Gauteng  79810846000~          7            2            1
11     2024 Gauteng  79911587000~          7            2            2
12     2024 Gauteng  79912914000~          7            1            1
13     2024 Mpumala~ 86810187000~          8            1            1
14     2024 Mpumala~ 86810539000~          8            2            1
15     2024 Mpumala~ 86910439000~          8            2            1
# i 12 more variables: fixed_internet_code <dbl>, mobile_internet_code <dbl>,
#   public_wi-fi_internet_code <dbl>, study_internet_code <dbl>,
#   work_internet_code <dbl>, library_internet_code <dbl>,
#   cafe_internet_code <dbl>, household_mobile_phones <dbl>,
#   household_smartphones <dbl>, computer_asset_code <dbl>,
#   geo_type_code <dbl>, household_weight <dbl>
```

## Graphs for our chosen datasets:

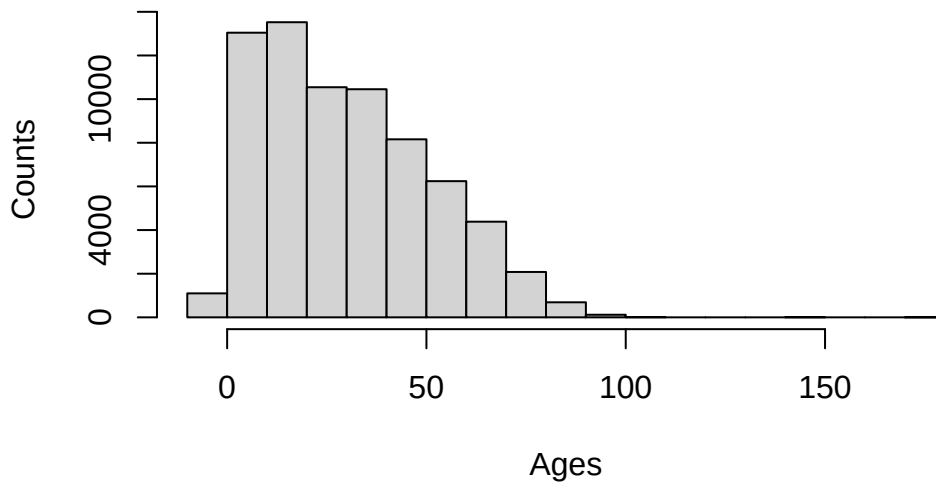
### Dataset 1 Graphs:

```
boxplot(D01$age, xlab = "Frequency", ylab = "Ages")
```



```
hist(D01$age, xlab = "Ages", ylab = "Counts")
```

## Histogram of D01\$age



For age we can see there are a couple of outliers at about 90, nothing to major though, we can also see that most people are around the age of 20, we also see some values below 0 that need to be dropped alongside the outliers

### Summary of D01 Ages

```
summary(D01$age)
```

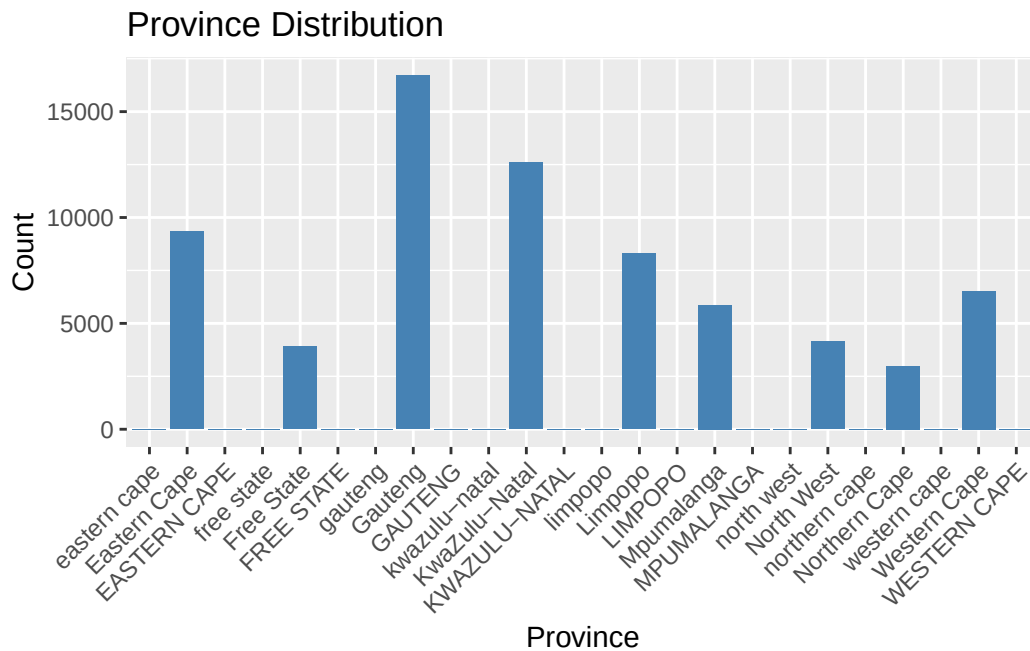
| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max.  | NAs |
|------|---------|--------|------|---------|-------|-----|
| -7.0 | 13.0    | 28.0   | 30.5 | 45.0    | 175.0 | 90  |

After a bit further investigation we see that the average age is 30 and that the majority of people are aged 28

### Province Distribution

```
D01 |>  
  ggplot(aes(x=province)) +  
  geom_bar(fill = "steelblue") +
```

```
labs(title = "Province Distribution",
     x = "Province",
     y = "Count") +
theme(axis.text.x = element_text(angle=45, hjust = 1))
```



This graph tells us that the spelling for all the different provinces are different.

### Finding Problem Province Naming

```
unique(D01$province)
```

```
[1] "Western Cape" "western cape" "WESTERN CAPE" "Eastern Cape"
[5] "eastern cape" "EASTERN CAPE" "Northern Cape" "northern cape"
[9] "Free State"   "free state"   "FREE STATE"   "KwaZulu-Natal"
[13] "kwazulu-natal" "KWAZULU-NATAL" "North West"   "north west"
[17] "Gauteng"      "GAUTENG"      "gauteng"      "Mpumalanga"
[21] "MPUMALANGA"  "Limpopo"      "LIMPOPO"      "limpopo"
```

By running unique province we prove this and we see there is a naming convention problem

## Fixing Provinces

```
D01 <- D01 |> mutate(province = str_to_title(province))
```

To fix it we mutate the province with `str_to_title` to normalize the capitalization of all the words

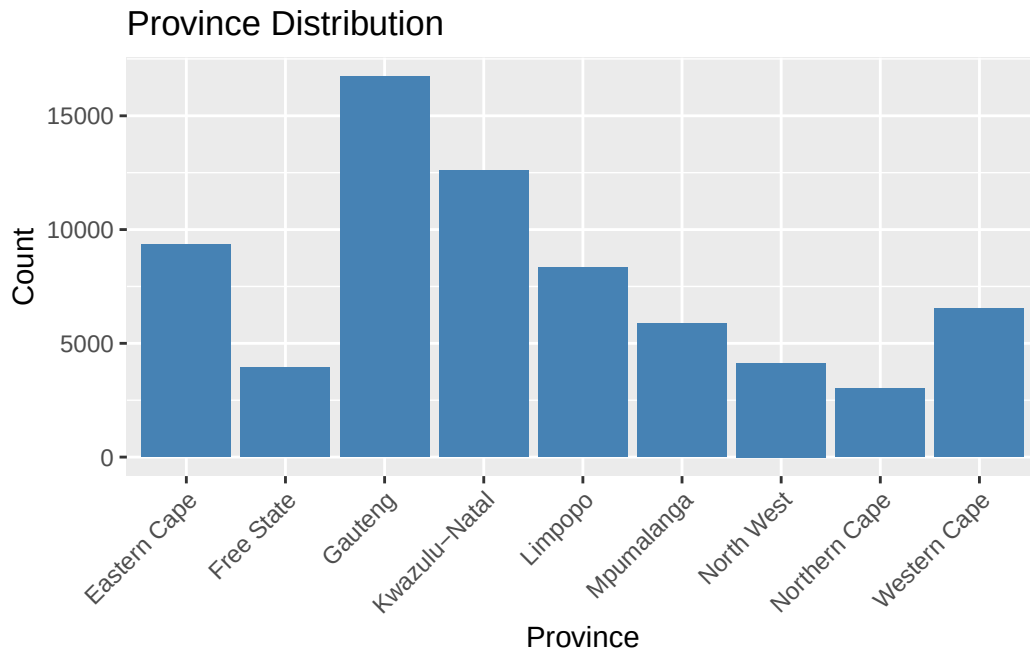
## Fixed Provinces

```
unique(D01$province)
```

```
[1] "Western Cape" "Eastern Cape" "Northern Cape" "Free State"  
[5] "Kwazulu-Natal" "North West"      "Gauteng"        "Mpumalanga"  
[9] "Limpopo"
```

## Province Distribution

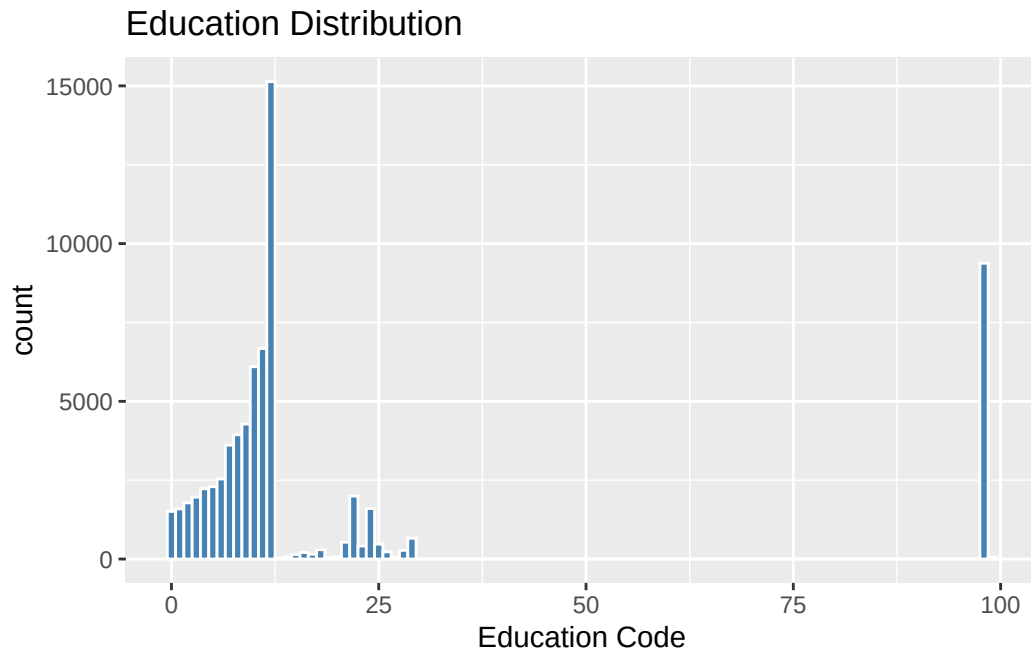
```
D01 |>  
  ggplot(aes(x=province)) +  
  geom_bar(fill = "steelblue") +  
  labs(title = "Province Distribution",  
        x = "Province",  
        y = "Count") +  
  theme(axis.text.x = element_text(angle=45, hjust = 1))
```



Our graph now looks nice, and we can see majority of the people live in Gauteng.

### Dataset 2 Graphs:

```
D02 |>
  ggplot(aes(D02$education_level_code)) +
  geom_histogram(bins = 50, binwidth = 1, fill = "steelblue", color="white") +
  labs(title = "Education Distribution",
       x = "Education Code",
       y = "count")
```



The rows the graphs has removed are the NA values that need to be dropped in the preprocessing stage.

From this graph however we can see that majority lies at around Code 10

### Showing Most Frequent Education Code

```
education_counts <- table(D02$education_level_code)
paste0("Most common code: ", names(which.max(education_counts)))
```

```
[1] "Most common code: 12"
```

```
paste0("Frequency of the code: ", max(education_counts))
```

```
[1] "Frequency of the code: 15136"
```

### Showing All Education Code Counts

```
data.frame(education_counts) |>
  arrange(desc(Freq))
```

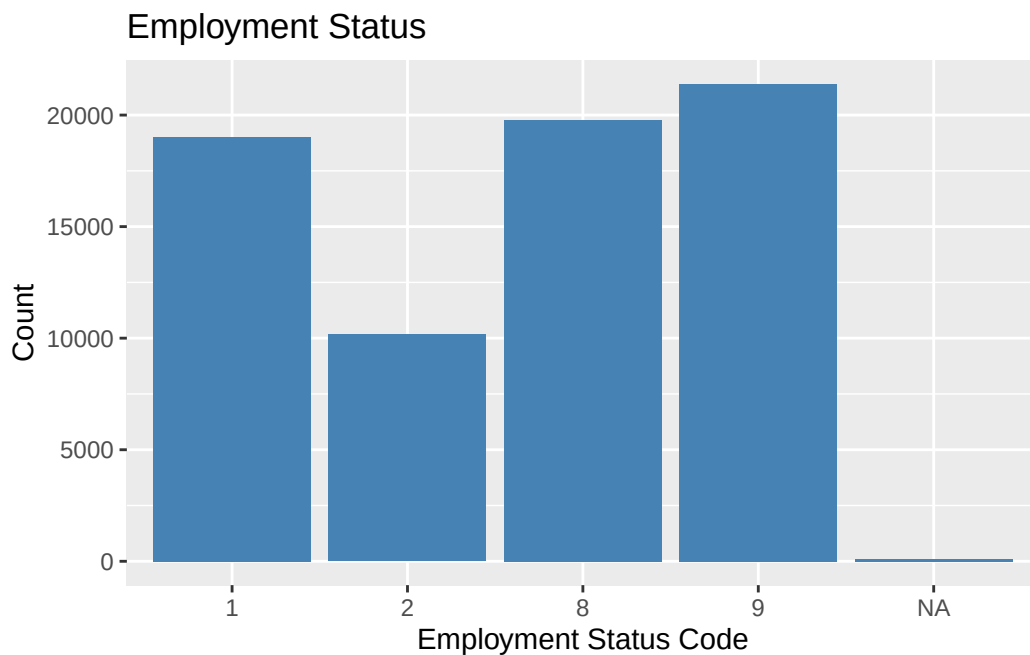
|    | Var1 | Freq  |
|----|------|-------|
| 1  | 12   | 15136 |
| 2  | 98   | 9378  |
| 3  | 11   | 6675  |
| 4  | 10   | 6098  |
| 5  | 9    | 4280  |
| 6  | 8    | 3940  |
| 7  | 7    | 3609  |
| 8  | 6    | 2539  |
| 9  | 5    | 2294  |
| 10 | 4    | 2228  |
| 11 | 22   | 1998  |
| 12 | 3    | 1963  |
| 13 | 2    | 1781  |
| 14 | 24   | 1604  |
| 15 | 1    | 1590  |
| 16 | 0    | 1510  |
| 17 | 29   | 660   |
| 18 | 21   | 533   |
| 19 | 25   | 472   |
| 20 | 23   | 413   |
| 21 | 18   | 296   |
| 22 | 28   | 276   |
| 23 | 26   | 228   |
| 24 | 16   | 206   |
| 25 | 17   | 151   |
| 26 | 15   | 144   |
| 27 | 20   | 70    |
| 28 | 27   | 67    |
| 29 | 14   | 66    |
| 30 | 99   | 53    |
| 31 | 19   | 51    |
| 32 | 13   | 33    |

After further exploration we see that code 12 is most common education which is Grade 12, with code 98 close behind it with 9378 records which means no schooling



### Dataset 3 Graphs:

```
D03 |>
  ggplot(aes(x = factor(employment_status1code))) +
  geom_bar(fill = "steelblue") +
  labs(title = "Employment Status",
       x = "Employment Status Code",
       y = "Count")
```



From this graph we can see Code 9 is the most with Code 8 being a close second Code 9 is unspecified and 8 is not applicable.

### Finding Most Frequent Employment Code

```
employment_codes_count <- table(D03$employment_status1code)
names(which.max(employment_codes_count))
```

```
[1] "9"
```

```
max(employment_codes_count)
```

```
[1] 21399
```

## Showing Employment Codes

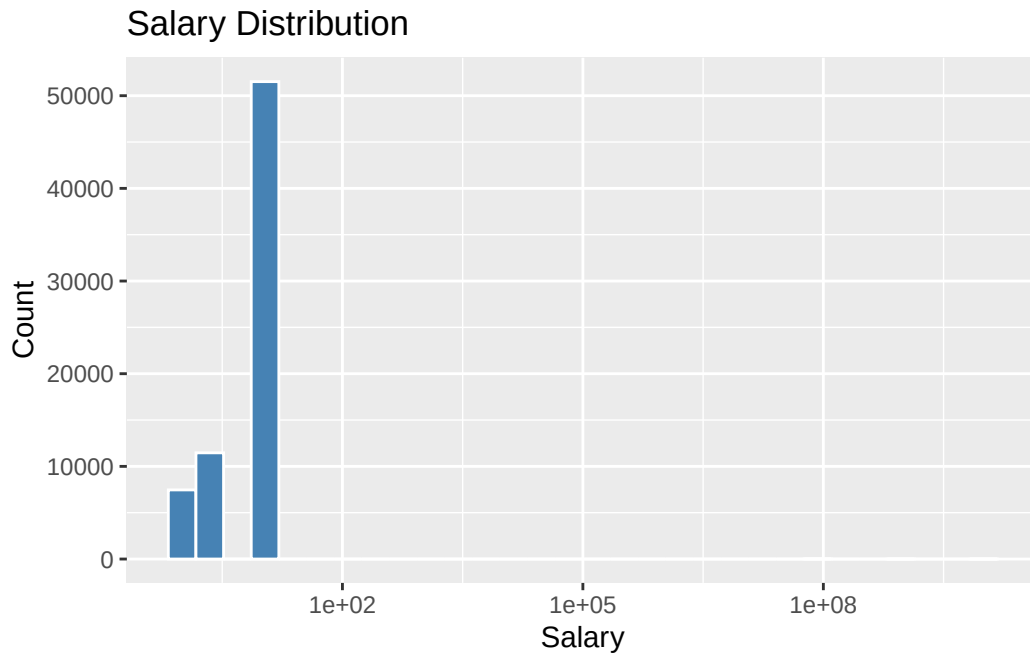
```
data.frame(employment_codes_count) |>  
  arrange(desc(Freq))
```

|   | Var1 | Freq  |
|---|------|-------|
| 1 | 9    | 21399 |
| 2 | 8    | 19784 |
| 3 | 1    | 19009 |
| 4 | 2    | 10168 |

We have now confirmed that Code 9 and 8 are the most with code 1 being a close second and code 2 being the least.

## Plotting Salaries

```
D03 |>  
  ggplot(aes(x = salary_amount_raw)) +  
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +  
  scale_x_log10() +  
  labs(title = "Salary Distribution",  
        x = "Salary",  
        y = "Count")
```



When we look at SalaryAmountRaw, these figures do not add up for actual salaries, the values are way too small

### Salary Summary

```
D03 |>
  filter(!lab_salary_raw %in% c(888888888, 999999999)) |>
  pull(lab_salary_raw) |>
  summary()
```

| Min. | 1st Qu. | Median | Mean  | 3rd Qu. | Max.    |
|------|---------|--------|-------|---------|---------|
| 1    | 2997    | 5257   | 10786 | 12000   | 1400000 |

The mean is larger than the median so a log would be appropriate to use on the graph

### Finding Unusable Salary Rows

```
placeholders <- c(888888888, 999999999)
D03 |>
  filter(lab_salary_raw %in% placeholders) |>
  nrow()
```

```
[1] 52145
```

### Finding Unusable Salary Row %

```
nrow(D03)
```

```
[1] 70460
```

```
52145 / nrow(D03) * 100
```

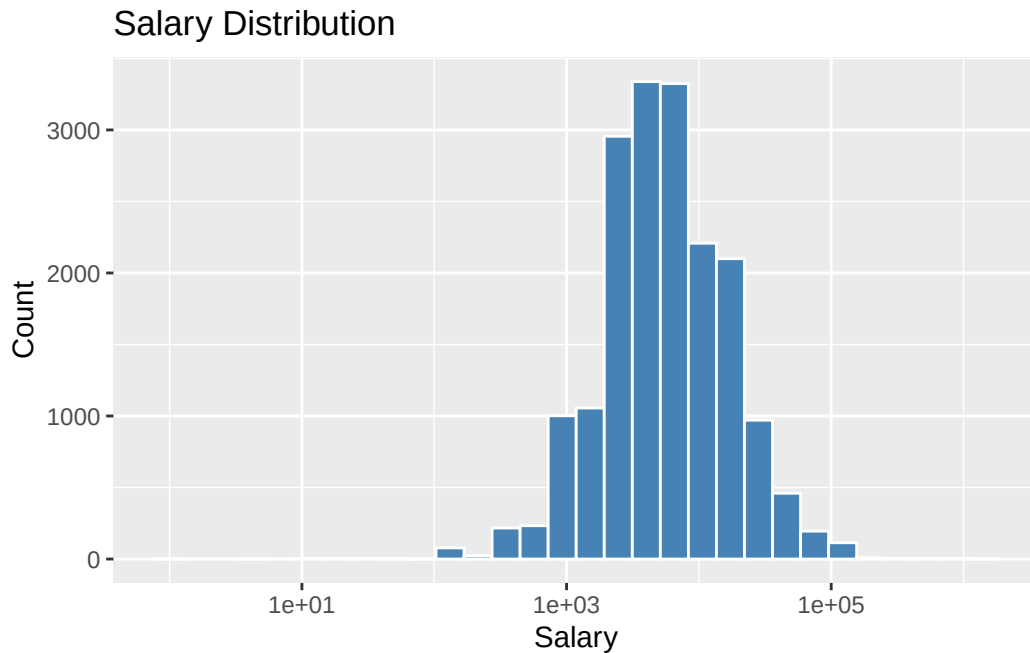
```
[1] 74.00653
```

Out of 70460 rows 52145 are unusable, which is approximately 74%

### Fixed Salary Graph

```
placeholders <- c(888888888, 999999999)

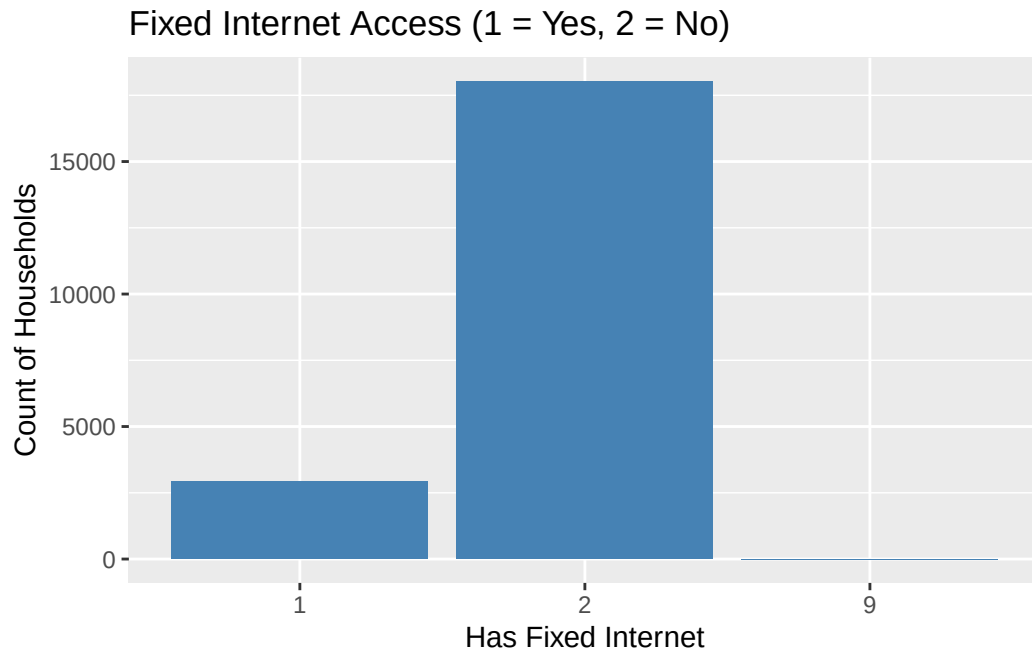
D03 |>
  filter(!lab_salary_raw %in% placeholders) |>
  ggplot(aes(x = lab_salary_raw)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  scale_x_log10() +
  labs(title = "Salary Distribution",
       x = "Salary",
       y = "Count")
```



We had to scale the values using log to get a better shaped graph and we had to remove the placeholder 888..etc and 999..etc so that the graph wouldnt get compressed too much

### Dataset 8 Graphs:

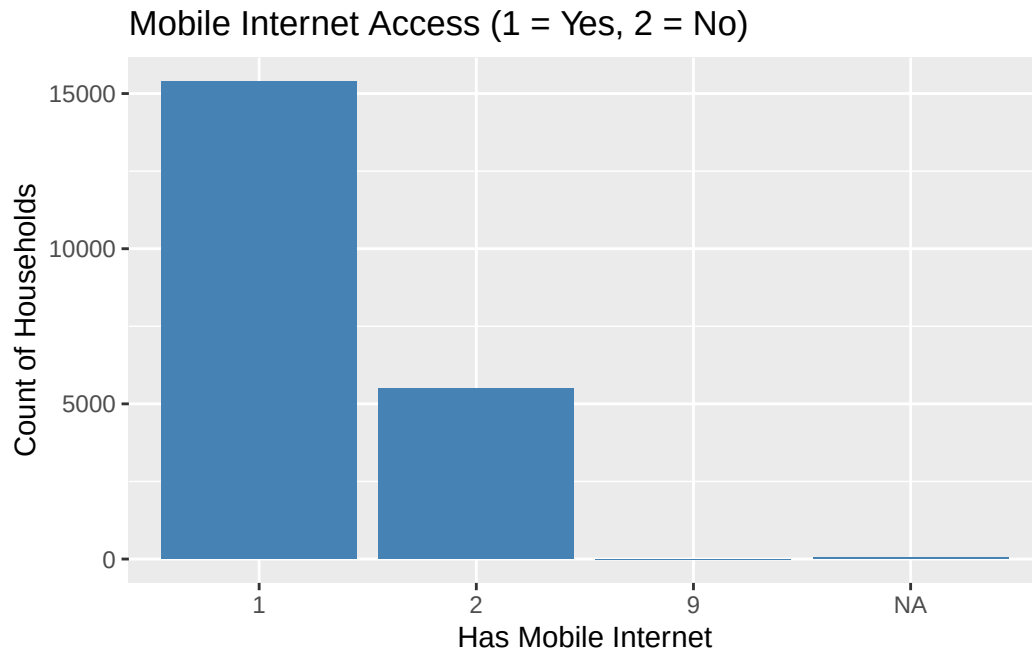
```
D08 |>
  ggplot(aes(x=factor(fixed_internet_code))) +
  geom_bar(fill = "steelblue") +
  labs( title = "Fixed Internet Access (1 = Yes, 2 = No)",
        x = "Has Fixed Internet",
        y = "Count of Households"
  )
```



We can see alot less people have fixed internet than those who do not have fixed internet.

### Mobile Internet Access

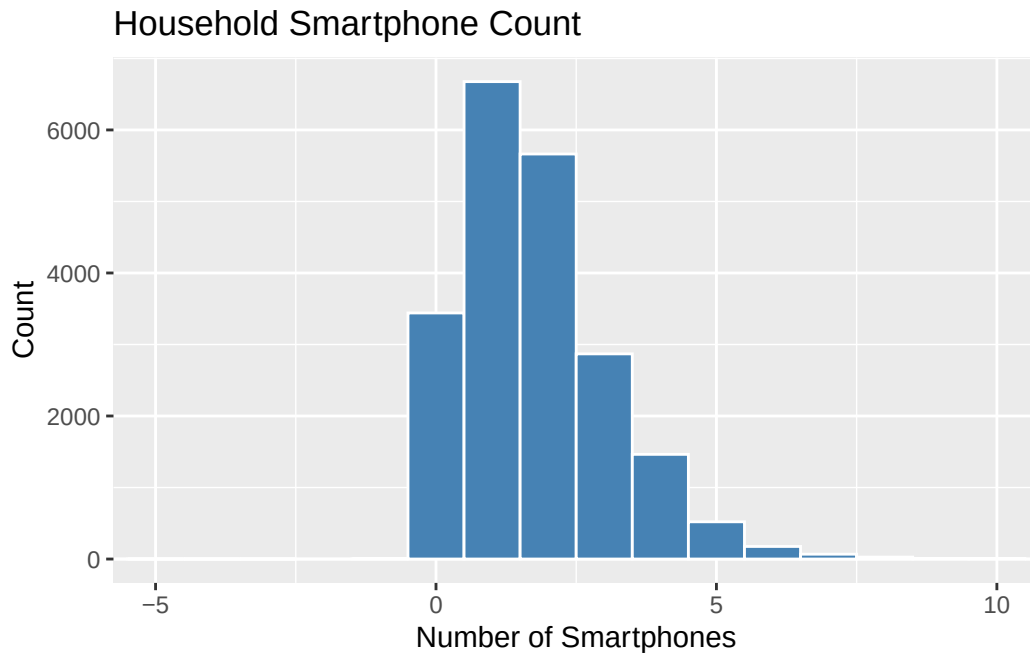
```
D08 |>
  ggplot(aes(x=factor(mobile_internet_code))) +
  geom_bar(fill = "steelblue") +
  labs( title = "Mobile Internet Access (1 = Yes, 2 = No)",
        x = "Has Mobile Internet",
        y = "Count of Households"
  )
```



We can see a lot more people have Mobile internet than those who do not have mobile internet.

### Household Smartphones

```
D08 |>
  ggplot(aes(x = household_smartphones)) +
  geom_histogram(binwidth = 1, boundary = 0.5, fill = "steelblue", color = "white") +
  coord_cartesian(xlim = c(-5, 10)) + # zooms visible window to the range -5 -> 10
  labs(title = "Household Smartphone Count",
       x = "Number of Smartphones",
       y = "Count")
```



## What to handle in preprocessing:

1. **Handle large values**, for example Household ID. It was read in as character to prevent the large numeric value from being rounded, this needs to be properly addressed in preprocessing.
2. **Missing Values**, D01 (115 missing), D02 (120 missing), D03 (100 missing), D08 (60 missing) all have missing values that need to be dropped.
3. **Duplicate Rows**, D01 (Na household ids create duplicate rows 23 duplicate rows), D02 (44 duplicate rows), D03 (40 duplicate rows), D08 has no duplicate rows nothing to be handled, duplicated rows need to be dropped.
4. **Invalid/out-of-range values**, D02,D03,D08 have invalid entries in some of their columns these need to be dropped, then the negative values for age in D01 also need to be dropped.
5. **Province naming inconsistencies**, other columns might also have invalid naming inconsistencies so run consistency validator on all columns to be safe.
6. **Remove extreme outliers**, columns like age have outliers way outside of the range which need to be removed. Perform scaling and outlier removal on all columns to be safe.



## Initial Hypotheses:

1. Higher education may be associated with a higher likelihood of employment.
2. Household internet / computer access may allow for education to lead to a better chance of employment.
3. Some provinces like gauteng might have better internet and more job opportunities, which means location might be the cause for higher education or better employment rate.
4. A higher income may be related to higher education and internet connectivity. However, the data here is weak since alot of records are missing, so this is more exploratory.